What Carries Deep-Noise Robustness in a Mamba-3 Block? Input-Independent Scans and Noise-Dependent Gating under Cross-Condition Shift

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Abstract

The robustness of Mamba-style selective state-space models is often attributed to input-dependent selectivity. We test this with controlled ablations of one Mamba-3 block design (instantiated as a four-layer stack) on bearing vibration data under cross-condition shift (XJTU-SY, one speed-and-load pair run in both directions, training and evaluation bearings disjoint), with matched-noise training and final-epoch macro-F1 under a fixed budget. All central contrasts were run with training and evaluation noise drawn independently. Blocks whose scan dynamics and read/write coefficients are made input-independent still outperform a time-invariant S4D (this is not a claim that selectivity contributes nothing or that full performance is preserved relative to the unfrozen block): on the original transition the fully frozen block leads by +21.3, +23.3, +20.3 pp at 0, −2 and −6 dB, and on the reverse transition by +27.5, +26.7, +21.1 pp. Removing the multiplicative output gate costs +5.4, +9.3, +32.6 pp in the fully frozen block, resolved only at −6 dB; only in this host and direction does the cost grow significantly with noise depth, and without its gate the block still exceeds S4D at 0 and −2 dB but not at −6 dB. Multiplicative gating exceeds an additive branch of identical size by +23.3 pp at −6 dB. On Paderborn bearings the gate-removal cost replicates when training and evaluation share bearings. Grafting a gate onto S4D, per-class results and Paderborn with disjoint bearings were run only

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under an earlier noise protocol and are descriptive. The evidence concerns one block design, one shift pair per dataset and five seeds.

Keywords: state-space models, Mamba, selectivity, gating, robustness, distribution shift

1. Introduction

Selective state-space models (SSMs) of the Mamba family [1, 2, 3] have rapidly become default sequence backbones, and a common narrative has formed around *why* they work: input-dependent selectivity—the ability to modulate state dynamics per token—is credited with filtering noise, adapting to distribution shift, and outperforming time-invariant predecessors such as S4D [4]. This belief motivates a growing family of "selective" architectures, yet recent evidence questions it: a systematic comparison on 59 clean multivariate time-series classification benchmarks (MONSTER and UEA) finds the *simpler, time-invariant* S4D consistently ahead of Mamba-family variants in accuracy and efficiency [5], and theoretical analyses identify expressivity regimes where Mamba is provably weaker [6]. Which account is right—and, more importantly, *what exactly should be credited* when a Mamba-style block does prove robust? This paper addresses the second question for one block and one testbed, with a controlled ablation chain whose decision thresholds were fixed before each grid was run. The testbed is bearing vibration diagnosis under operating-condition drift and matched noise, in which the robustness gap is large (12–28 pp between S4D and the frozen Mamba-3 blocks in the independent-noise results), the evaluation labels are identical across all cells of each transition (hash-verified), and the block can be dissected.

Failure to identify the source of an empirical gain is a recurring weakness of machine-learning scholarship [7]. Our approach is deliberately *attributional* rather than *methodological*: we propose no new architecture and claim no state-of-the-art. We take a regime in which a Mamba-style block shows a large, stable robustness advantage over a time-invariant SSM, and ask which ingredient earns it, in three steps with adjudication thresholds fixed before each grid was run:

1. **Freeze the scan.** The input-dependent parts of Δ, A, B, C are replaced by learned constants; in a second arm the trapezoidal weight and rotation angles are as well, so that the scan's dynamics and read/write

coefficients no longer depend on the input. The value stream and the gate input remain input-dependent (Section 3.2).

2. **Remove the gate.** On the frozen base the multiplicative SiLU gate is removed, with the other components and the budget unchanged.
3. **Replace or transplant the gate.** In place, the gate is replaced by an additive branch of identical size and initial weights; across hosts, it is grafted onto plain S4D with an additive control.

Steps 1 and 2 and the in-place replacement were tested with training and evaluation noise drawn independently (Section 4); the graft onto S4D, the per-class analysis, the stem and A ablations and the disjoint-bearing Paderborn setting were run only under an earlier protocol in which the two noise streams were shared by window index, and are reported in Appendix D as descriptive evidence (Table 3). The outcome is a conditional pattern. Making the scan input-independent does not remove the block's advantage over S4D, on either transition (+21.3, +23.3, +20.3 pp and +27.5, +26.7, +21.1 pp for the fully frozen block). Removing the gate costs +9.4, +15.7, +27.0 pp in the partially frozen block on the original transition, resolved only at −6 dB, and 19.3, 22.7, 22.0 pp on the reverse transition, resolved at every level. Only in the fully frozen block on the original transition does the gate-removal cost grow significantly with noise depth: without its gate that block still exceeds S4D at 0 and −2 dB but not at −6 dB. An additive branch of identical size does not recover the gate's effect at −6 dB in either host, although the pre-specified rescue test is not assessable. On Paderborn bearings the gate-removal cost replicates when training and evaluation share bearings.

Contributions. (i) A layered freezing experiment on a Mamba-3 block, down to a scan with input-independent dynamics and read/write coefficients (verified at the kernel interface), in which the advantage over S4D survives on both transitions of one dataset under independent training and evaluation noise. (ii) Same-host tests of the output gate (removal and in-place additive replacement) in a partially and a fully frozen block, with a descriptive analysis of how the gate's contribution changes with noise depth. As supporting material, the study reports a coverage table by protocol, the correction of an initial noise-protocol defect, and a released pre-specification and review trail, including the incidents in which automated adjudication departed from the pre-specified criteria (Appendix C).

2. Background and Related Work

2.1. Time-invariant and selective SSMs

Diagonal structured SSMs (S4D) [4], a simplification of S4 [8] whose initialisation derives from HiPPO [9], evolve a linear state with input-independent parameters (A, B, C, Δ), acting as a bank of learned linear filters. Related alternatives to attention [10] include S5, a multi-input multi-output linear SSM [11], the linear recurrent unit [12], and Hyena, which interleaves implicitly parametrised long convolutions with data-controlled gating [13]. Mamba-style blocks [1] make Δ, B, C functions of the input token (selectivity), wrap the scan in a block containing a short causal convolutional stem, a multiplicative SiLU gate on a parallel branch, and residual/normalisation structure; the concrete block used throughout this paper follows the Mamba-3 formulation [14] (see also Mamba-2 [15]). The narrative crediting selectivity for robustness draws on theory showing input-dependent dynamics can represent discontinuity-sensitive projections and counteract memory decay [16], and on [17]. We treat these as *hypotheses to be tested*, not established attributions: the theoretical results concern representability, which does not by itself determine which component earns empirical robustness in a trained block.

Multiplicative gating is not specific to Mamba. It has been used in recurrent networks since the LSTM [18], in highway networks [19], in gated convolutional language models [20] and their transformer variants [21], and as channel-wise recalibration in convolutional networks [22]. The gate of a Mamba-style block applies the SiLU activation [23, 24] to a parallel branch and multiplies the result onto the scan output; this is the operation whose role the present paper isolates.

2.2. The unresolved comparison

Empirically the picture is contested. A systematic comparison on 59 clean multivariate time-series classification benchmarks (MONSTER and UEA) finds S4D-based models superior to Mamba variants [5]; that study does not include additive noise or a shift of operating condition between training and evaluation, which is the regime examined here; theoretical work identifies long-range regimes where Mamba is weaker than S4D [6], and copying [25], state-tracking [26] and associative-recall [27] studies delimit tasks on which fixed-state sequence models lag attention. In vision, where Mamba backbones have been adopted [28], it has been hypothesised that Mamba is

not necessary for image classification (discussed below). Meanwhile, application literatures—including machine-condition monitoring—contain many Mamba-hybrid methods reporting noise robustness [29, 30, 31, 32, 33], but none, to our knowledge, isolates *which component* of the block is responsible under controlled conditions. Our clean-data results agree with the S4D-superiority finding (S4D matches or slightly exceeds the Mamba-style block on clean data). Whether the disagreement in the literature reflects a regime difference is not settled by a single testbed; what we can report is that, in the drift $\times$ noise regime studied here, the Mamba-style block's advantage over S4D is not removed by freezing the classical selectivity terms.

Relation to the closest studies. Three studies are closest in question or method, and the present work differs from each in a specific way. (i) MambaOut [34] stacks Mamba blocks with their SSM token mixer removed, leaving a gated convolutional block, and reports that on ImageNet classification such models surpass visual Mamba models; it asks, for whole vision models, whether the SSM is needed. Here the selective-scan layer is kept in every arm, its input-dependent parameters are replaced by learned constants step by step (Section 3.2), and the output gate is then removed or replaced by an additive branch inside the same host, on one-dimensional vibration signals under matched noise and operating-condition shift; nothing is claimed about vision. (ii) The 59-dataset comparison [5] compares complete S4D-based and Mamba-based architectures on clean benchmarks; the differences are the evaluation regime (noise and condition shift) and the level of intervention (components within one block family rather than complete models). (iii) The selectivity theory [16, 6] concerns what input-dependent dynamics can represent; the experiments here are empirical and do not test representability, so they neither confirm nor contradict those results. What this paper adds is therefore bounded: a component-level attribution inside one block design, in one noisy cross-condition regime.

2.3. Testbed rationale

We use bearing vibration diagnosis under cross-condition drift (speed and load both change) as the testbed, for three properties that matter to the attribution question rather than to the application: (a) the distribution shift is physically generated (a change of rotation speed and radial load between recorded conditions), not simulated; (b) fault signatures are sparse transients—precisely the discontinuous structures selectivity theory predicts selective SSMs should excel at [16], which makes it a plausible arena for the

selectivity hypothesis; and (c) the robustness gap between the Mamba-style block and S4D is large (12–28 pp in the independent-noise results), giving the attribution chain high signal. Bearing diagnosis with machine learning is reviewed in [35, 36]; the Case Western Reserve University data are the best-known benchmark [37] and have been used to study noise and load changes with convolutional networks [38], while open-source benchmark studies stress comparable protocols [39]. Operating-condition shift is usually addressed with domain adaptation such as adversarial training [40]; we use none, because the question is which component of one block carries robustness, not how to close a domain gap. Additive noise at a fixed SNR is applied in the manner of corruption benchmarks [41]. The testbed is two-class (details in Section 3.1); the scope of all claims is bounded accordingly (Section 6).

3. Experimental Protocol

3.1. Testbed, arms, and budget

Vibration windows from the XJTU-SY run-to-failure dataset [42] are split by operating condition: models train on one operating condition (37.5 Hz, 11 kN) and evaluate on another (40 Hz, 10 kN), so speed and load both change (cross-condition), with additive white Gaussian noise applied at matched SNR to training and evaluation ({clean, +10, +6, 0, −2, −6, −10 dB}). The task is binary fault classification (inner-race vs. outer-race dominant classes; class imbalance makes IR the hard class, Appendix D.10). All arms share an identical training budget (50 epochs, 5 seeds; the per-class grid of Appendix D.10 used 20 epochs), identical splits, and identical evaluation labels, verified by a hash of the label vector: every cell on the original transition carries `eval_sha` = `6c20b367`… (Appendix A), recomputed independently during adjudication; the reverse transition and the two Paderborn settings carry their own label hashes (`93424834`…, `53d565d9`… disjoint, `addc1938`… shared). The hash certifies the label vector, not the input windows. In total the study comprises 865 cells across five pre-specified grids of the original design (baseline grid 475; frozen-selectivity 105; component ablation 175; graft 35; per-class 75), run under the earlier noise protocol, and follow-up grids add 645 cells: 210 under independent training and evaluation noise (Sections 4.1–4.3) and 435 under the earlier protocol (Appendix D); Table 3 gives the coverage. Training and evaluation bearings are disjoint in each direction: Bearing2_1, 2_2, 2_4, 2_5 at 37.5 Hz/11 kN and Bear-

ing3_1, 3_3, 3_4, 3_5 at 40 Hz/10 kN; the reverse transition swaps their roles.

Composition of the data. Training uses the bearings of the 37.5 Hz/11 kN condition whose failure mode is inner race or outer race (Bearing2_1 is labelled IR; Bearing2_2, 2_4 and 2_5 are labelled OR); evaluation uses those of the 40 Hz/10 kN condition (Bearing3_3 and 3_4 are IR; Bearing3_1 and 3_5 are OR). Bearings with cage or mixed failure modes are excluded. Windows are non-overlapping segments of 2048 samples of the horizontal channel taken from the fault phase, which is delimited by a kurtosis/RMS onset criterion with a last-15% fallback; the evaluation set holds 3,504 OR and 1,744 IR windows, the training set 4,624 OR and 624 IR windows. Two consequences matter for interpretation. The IR class is learned from a single training bearing, so class-level results may reflect bearing-specific generalisation. And the five seeds resample initialisation, batch order and noise but not bearings: there are four independent bearings on each side, and none of the intervals reported here captures bearing-level uncertainty. One follow-up grid (Section 4.2) reverses the transition (40 Hz/10 kN → 37.5 Hz/11 kN). It uses the same eight bearings with their roles swapped, so it tests the other direction rather than independent bearings; its evaluation set holds 4,624 OR and 624 IR windows, all IR windows coming from Bearing2_1, and its label hash is `93424834`…. The 35 Hz/12 kN condition has no inner-race bearing and cannot form a binary pair.

The principal arms are summarised in Table 1, and the configuration that the code fixes for all of them in Table 2.

Table 1: Principal experimental arms and parameter counts.

Arm	Description	Params
BM3	Mamba-style block stack (conv stem k=7 → selective scan → SiLU gate → projection), with and without an auxiliary physics loss (Section 3.4)	177,938
BM3-frozen	BM3 with the input-dependent parts of Δ, A, B, C replaced by learned constants; trapezoidal weight, rotation angles, stem, gate and the remaining components unchanged (Section 3.2)	112,410
S4D	Parameter-matched diagonal time-invariant SSM (LayerNorm; no BatchNorm anywhere in either SSM arm)	100,162
S4D-wide	S4D widened to 1.75× parameters (capacity control)	178,498
frozen−conv	Single-component removal on the frozen base: conv stem (Appendix D.3)	112,026

Table 1: (continued)

Arm	Description	Params
frozen−gate	Single-component removal on the frozen base: SiLU gate (Appendix D.3)	79,642
frozen−A	Single-component removal on the frozen base: heavy-tailed A parameterisation (Appendix D.3)	112,410
frozen−trap	Frozen base with the trapezoidal weight also made a learned constant (Section 3.2)	111,906
frozen−rot	Frozen base with the rotation angles also made learned constants	104,346
frozen−all	Frozen base with both also constant: scan dynamics and read/write coefficients input-independent (value stream and gate still input-dependent)	103,842
frozen−all−gate	frozen−all with the gate rows removed from the input projection (Section 4.1)	71,074
frozen−all+add	frozen−all with $y \odot \mathrm{SiLU}(z)$ replaced by $y + \mathrm{SiLU}(z)$; same parameters and initial weights	103,842
S4D+gate	Plain S4D plus the SiLU gate branch only (graft, Appendix D.4)	116,802
S4D+gate (wm)	Graft rebuilt in the donor's shape: per layer x, z projected to width 128, S4D at width 128 (state 64, same SSM budget as the host), $W_{\mathrm{out}}(y \odot \mathrm{SiLU}(z))$ (Appendix D.5)	198,978
S4D+branch	Same modules, parameters and initial weights, combined additively: $W_{\mathrm{out}}(y + \mathrm{SiLU}(z))$ (control)	198,978
frozen+add	BM3-frozen with the gate replaced in place: $W_{\mathrm{out}}(y + \mathrm{SiLU}(z))$ instead of $W_{\mathrm{out}}(y \odot \mathrm{SiLU}(z))$; same modules, parameters and initial weights (Appendix D.6)	112,410
CNN/TCN baselines	Conventional baselines for context	—

3.2. What the frozen arm freezes, and what the checks verify

In the Mamba-3 layer used here, one input projection emits the gate and value streams, the scan inputs B and C, the step-size and decay terms (Δ and A, through input-dependent additive terms), a trapezoidal weight and rotation angles. The frozen arm removes the input-dependent parts of Δ, A, B and C and replaces them with learned constants broadcast over time (for Δ the static bias of the original layer is kept; for A a new constant parameter plays the role of a static log-decay, passed through the layer's own heavy-tailed clamp). The removed projections are deleted, so no parameter or gradient can leak. **The trapezoidal weight and the rotation angles remain functions of the current input**, as do the stem and the output

Table 2: Configuration fixed by the training code for all arms (differences between arms are noted). OR: outer race, IR: inner race.

Setting	Value
Signal	Horizontal channel at 25.6 kHz; non-overlapping 2048-sample windows (16 per 32,768-sample file) from the fault phase.
Fault phase	Starts at the earlier of (a) the first file at which the rolling mean over 5 files of the per-file kurtosis exceeds 5.0 and (b) the first file at which the rolling mean of the per-file standard deviation exceeds twice its baseline (mean over the first $\min(20, N/5)$ files, N files in the run); if neither occurs, the last 15% of the files.
Scaling and noise	Each window is z-scored, then white Gaussian noise is added at the stated SNR to training and evaluation windows.
Network	$d_{\text{model}} = 64$, 4 layers, state size 128, LayerNorm; input stem Conv1d($1 \to 64$), stride 2, padding 3, kernel 7 in the Mamba-3 arms and 8 in S4D; Mamba-3 expansion 2 ($d_{\text{inner}} = 128$).
Optimisation	AdamW, learning rate 3×10^{-4}, weight decay 10^{-4}, cosine annealing to zero over 50 epochs, batch size 64 (evaluation 128), gradient-norm clipping at 1.0.
Sampling	Training windows are drawn with replacement with probability inversely proportional to class frequency, as many draws per epoch as training windows. With 4,624 OR and 624 IR training windows (original transition) the expected class shares are equal instead of 11.9% IR, so IR windows are drawn about 4.2 times as often as their natural frequency. This holds for every arm; its interaction with the gate was not tested.
Loss and seeds	Cross-entropy only in all attribution arms. Seeds 0–4 set initialisation, sampler draws and noise; they do not resample bearings.

gate. The frozen arm is therefore not a time-invariant scan: in the Mamba-3 formulation [14] the input-dependent rotation and trapezoidal weighting also enter the state update. Appendix D.1 is accordingly a statement about the classical selectivity terms Δ, A, B, C only. Three further arms close that gap by the same device: *frozen−trap* replaces the trapezoidal weight by a learned per-head constant, *frozen−rot* replaces the rotation angles by learned constants (so the rotation becomes a learned position-only phase rather than a function of the current input), and *frozen−all* does both, so that the scan's dynamics and read/write coefficients (Δ, A, B, C, trapezoidal weight, rotation angles) no longer depend on the input. The value stream x and the gate input z are still computed from the input, as they must be for the layer to carry any signal; the rotation becomes a learned position-dependent

phase. In each case the corresponding `in_proj` output slice is deleted. Appendix D.2 reports them. For these arms the construction check adds the decisive probe: the tensors handed to the scan are captured for two different inputs through the same module, and a tensor the arm claims to have frozen is required to be bitwise identical across them while a tensor left selective is required to differ (all three arms pass). No kernel modification is needed, because the scan interface accepts these tensors regardless of their origin.

Single-variable discipline was checked by key-level state-dictionary diffs between two models built under the same seed. For the frozen arm the diff verifies that every non-target tensor has the same name and shape as in BM3; for the component-ablation and graft arms it additionally compares the values of non-target tensors at initialisation. Every differing key is mapped to an entry in an architecture map. We do not claim byte-level identity of non-target tensors in the frozen arm, and trained weights naturally differ between arms; the checks concern how the arms were constructed.

3.3. Reporting rule

Every cell is trained for a fixed budget of 50 epochs with AdamW [43] and a cosine learning-rate schedule [44] that anneals to zero, and the macro-F1 reported in the main text is that of the *final epoch*, a rule that uses no evaluation data to choose a model. The original grids recorded instead the maximum over epochs of macro-F1 on the evaluation split (the harness defines no separate validation split; the choice of model-selection rule can change conclusions in domain-generalisation benchmarks [45]), and the thresholds pre-specified for those grids were written against that rule. The arms of the original transition were therefore re-run on the adjudication band (0, −2, −6 dB) with the original training code, the same evaluation fingerprint and per-epoch evaluation (final-epoch re-run, earlier noise protocol; Appendix D.9). This final-epoch re-run and the independent-noise re-run of Section 4 are different sets of cells with different coverage (Table 3). Best-epoch values are a sensitivity analysis. Levels outside the band (clean, +10, +6, −10 dB) and the per-class grid were not re-run; values there are best-epoch and auxiliary. Because the final epoch of the same evaluation condition has been inspected repeatedly during this study, final-epoch values are not a fresh confirmatory test either.

3.4. Confound elimination

Before component attribution, three candidate confounds were examined on the same grid. **Capacity (earlier protocol):** S4D-wide (1.75× parameters) tracks S4D (S4D-wide − S4D: −0.4, −1.0, +0.4 pp at 0, −2, −6 dB under final-epoch reporting, all intervals including zero), while BM3-frozen exceeds S4D-wide by +15.9, +20.8, +16.0 pp with intervals excluding zero at all three levels, so this particular increase in S4D's parameter count does not explain the gap; the control does not address the change in channel mixing that gate removal also entails. **Normalisation:** both SSM arms use LayerNorm and contain no BatchNorm [46]; the arm that degrades under noise is the one *without* any batch statistics, which argues against BatchNorm statistics mismatch as the explanation. Other normalisation inside the Mamba-3 layer was not varied. **Noise streams:** in the original harness the noise added to window i was drawn from a generator keyed on (seed, i), and training and evaluation used the same seed, so window i of the training set and window i of the evaluation set carried the same base noise vector. Per-index label agreement between the two sets is 0.549 on XJTU-SY, below the 0.628 expected by chance, which excludes only the simplest route (memorising the training noise attached to each index); on Paderborn it is 0.999 (disjoint) and 1.000 (shared). All central contrasts were therefore re-run with the stream keyed on (seed, split, i), which makes training and evaluation noise independent while all arms with a given seed share one evaluation noise bank (Section 4.1); the reverse transition was re-run in the same way (Section 4.2). The component ablations of the stem and the state parameterisation, the grafts, the additive substitution on the reverse transition, the per-class grid and the disjoint-bearing Paderborn setting were not re-run; they carry the earlier protocol in their captions and are reported in Appendix D. In the arms that were re-run the means moved by at most 4 pp and no contrast changed direction; this is an observation about those arms, not a guarantee for the ones that were not re-run, and the earlier-protocol results are weighted accordingly. **Auxiliary physics loss:** an exploratory kinematics-alignment regulariser carried by the original model contributes −0.1 pp pointwise (curves coincide); all attribution arms are therefore built on the pure cross-entropy variant.

3.5. Pre-specification and adjudication

Every adjudicated question in this study had numeric decision thresholds written to a task card and to disk before the corresponding grid was

launched. This is internal pre-specification, not external registration: the supporting evidence is local file modification times and run-directory names (released, Appendix A), which are not tamper-evident and are not preserved by a repository checkout, so the trail lets readers inspect the order of events but cannot prove it. The exception is the follow-up grids from the width-matched graft onwards (Sections 4.1–4.3 and Appendix D.5–Appendix D.8): their thresholds, and for some of them the title decision that depends on their outcome, were committed to the public repository before the first cell was run (including the smoke check), which gives those grids an external time-stamp. Later questions (component ablation, graft) were specified after earlier grids had been seen, as is natural in a staged design. Adjudication was mechanical (an analysis pass applies the thresholds and a separate reviewer pass re-computes the aggregates); the incidents in which automated adjudication departed from the pre-specified criteria, and the human adjudication gates, are described in Appendix C. In the grids of the original design the adjudication criteria were effect-size thresholds, and paired intervals and tests were reported but not used as criteria. In the later grids (Section 4) the labels *holds*, *weakened* and *does not hold* are defined by whether the paired 95% interval excludes zero at two, one or none of three levels, so interval position is part of the adjudication there. A single repeat of one 2-epoch configuration in a separate process differed by ~ 0.7 pp macro-F1 (non-deterministic kernel reductions; Appendix B); this is an observed repeat difference, not an estimated variability bound, and arm-level contrasts are judged against seed-level variability (2–9 pp).

4. Results

Throughout, "deep noise" denotes the pre-specified adjudication band $\{0, -2, -6\,\mathrm{dB}\}$; -10 dB is reported only in the earlier-protocol appendix and excluded from adjudication (the time-invariant baseline is near chance there). All values are macro-F1 (%), mean over 5 seeds, at the final epoch of the 50-epoch budget (Section 3.3). **The main text reports only results obtained with independent training and evaluation noise** (Sections 4.1–4.4). Results obtained while training and evaluation noise were shared by index (Section 3.4) are collected in Appendix D and summarised, with their evidence level, in Section 4.5. Table 3 shows which operation was run under which protocol on which host and transition. Figure 1 summarises the chain.

Table 3: Coverage of the study by protocol. I: run with independent training and evaluation noise (main text). E: run under the earlier protocol (Appendix D). –: not run. P: partially frozen host (BM3-frozen); F: fully frozen host. "Freeze" is the contrast with S4D (BM3-frozen − S4D for P, frozen−all − S4D for F); "gate removal" and "additive substitution" are applied inside the host. Graft, stem/A removal and per-class recall have no fully frozen counterpart.

	XJTU forward		XJTU reverse		PU shared		PU disjoint	
Operation	P	F	P	F	P	F	P	F
Freeze (vs S4D)	I	I	I	I	I	–	E	E
Gate removal	I	I	I	–	I	–	E	–
Additive substitution	I	I	E	–	–	–	E	–
Graft onto S4D (with additive control)	E	–	E	–	–	–	–	–
Stem and A removal	E	–	–	–	–	–	–	–
Per-class recall	E	–	–	–	–	–	–	–

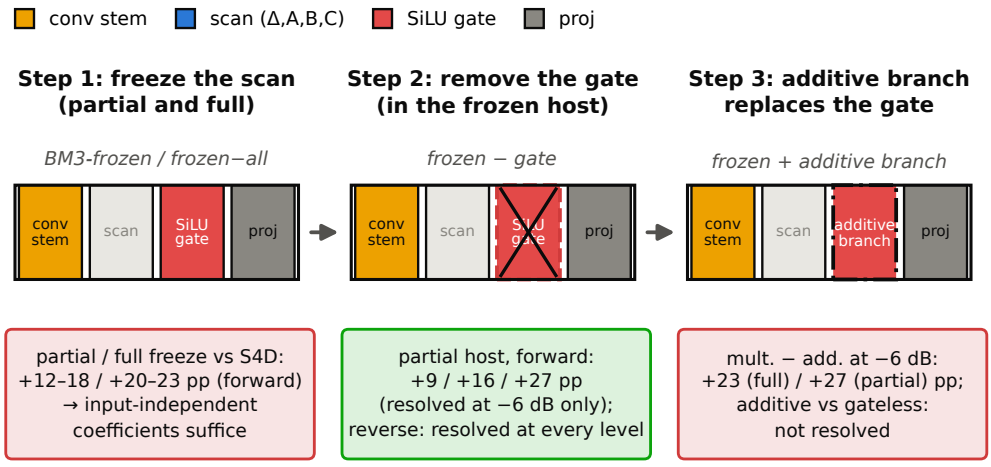


Figure 1: The three-step chain. Steps 1 to 3 are tested under independent training and evaluation noise (Sections 4.1–4.4); step 3 is the in-host replacement of the multiplicative gate by an additive branch, on the partially and the fully frozen host. The graft of the gate onto S4D was run only under the earlier noise protocol and is an exploratory branch in Appendix D. Numbers in the boxes are final-epoch differences in percentage points.

4.1. Original transition under independent noise

A pre-specified grid of 150 cells (thresholds, labels and writing consequences committed to the public repository before the first cell) re-ran the central arms with independent training and evaluation noise (Section 3.4): 105 cells on XJTU-SY (the seven arms of Table 4, three levels, five seeds) and 45 on Paderborn shared bearings (three arms, Section 4.3); everything else is unchanged. Each contrast is labelled *holds* if at least two of the three levels have a positive mean with an interval excluding zero and none is reversed, *weakened* if exactly one, *does not hold* if none. Relative to the earlier protocol the arm means moved by −3.4 to +3.9 pp over the eight arm-by-setting pairs that have an earlier-protocol counterpart (five XJTU-SY arms and three Paderborn arms); no contrast changed direction.

Table 4: Independent training and evaluation noise, original XJTU-SY transition (37.5 Hz/11 kN → 40 Hz/10 kN). Upper block: macro-F1 (%) at the final epoch, mean over 5 seeds. Lower block: paired differences with 95% t-intervals and the number of seeds (of 5) with a positive difference, and the pre-specified label. ‡Added post hoc, not pre-specified.

Arm / contrast	0 dB	−2 dB	−6 dB	label
BM3-frozen	80.3	86.4	87.0	
frozen−all	89.5	91.5	88.9	
S4D	68.3	68.3	68.6	
frozen−gate	70.9	70.7	60.0	
frozen+add	71.3	71.3	60.1	
frozen−all−gate	84.1	82.2	56.3	
frozen−all+add	84.9	83.6	65.6	
frozen−all	+21.3	+23.3	+20.3	
− S4D	[14.6, 27.9] (5/5)	[20.2, 26.4] (5/5)	[17.9, 22.8] (5/5)	holds
BM3-frozen	+12.1	+18.1	+18.4	
− S4D	[0.9, 23.2] (5/5)	[9.8, 26.4] (5/5)	[12.7, 24.0] (5/5)	holds
BM3-frozen	+9.4	+15.7	+27.0	
− frozen−gate	[−0.5, 19.4] (5/5)	[−0.8, 32.1] (4/5)	[7.0, 46.9] (4/5)	weakened
frozen−all	+5.4	+9.3	+32.6	
− frozen−all−gate	[−3.8, 14.6] (3/5)	[−0.5, 19.1] (5/5)	[28.9, 36.3] (5/5)	weakened
frozen−all−gate	+15.9	+14.0	−12.3	
− S4D	[6.9, 24.8] (5/5)	[2.5, 25.4] (4/5)	[−17.5, −7.0] (0/5)	descriptive
BM3-frozen	+9.0	+15.1	+26.8	
− frozen+add	[−0.8, 18.9] (5/5)	[−0.9, 31.0] (4/5)	[6.8, 46.9] (4/5)	descriptive‡
frozen−all	+4.6	+7.9	+23.3	
− frozen−all+add	[−4.0, 13.2] (3/5)	[1.0, 14.8] (5/5)	[9.0, 37.5] (5/5)	descriptive‡
frozen+add	+0.4	+0.6	+0.1	
− frozen−gate	[−0.6, 1.4] (4/5)	[−0.1, 1.2] (5/5)	[−0.4, 0.7] (3/5)	descriptive
frozen−all+add	+0.8	+1.4	+9.3	
− frozen−all−gate	[−1.5, 3.1] (2/5)	[−1.9, 4.7] (2/5)	[−7.2, 25.8] (5/5)	descriptive

Freezing. Both freeze contrasts hold. The fully frozen block exceeds S4D by +21.3, +23.3, +20.3 pp, in every seed at every level; the partially frozen block exceeds it by +12.1, +18.1, +18.4 pp.

Gate removal. Removing the gate from the partially frozen block costs +9.4, +15.7, +27.0 pp; the contrast is *weakened* because only the −6 dB interval excludes zero. In the fully frozen block it costs +5.4, +9.3, +32.6 pp, again resolved only at −6 dB. Without its gate the fully frozen block still exceeds S4D by +15.9 and +14.0 pp at 0 and −2 dB (intervals excluding zero) and falls 12.3 pp below it at −6 dB (interval $[-17.5, -7.0]$, all seeds). The gate is therefore not required for a positive advantage over S4D at 0 and −2 dB, but it is required at −6 dB. At the two milder levels the removal cost is +5.4 and +9.3 pp with intervals that include zero, so a contribution of the gate there is neither demonstrated nor excluded. The pre-specified condition for a combined attribution of the fully frozen block's advantage to the gate (gate removal holds and the additive substitution is not rescued) is not met, and we make no such claim.

Multiplicative versus additive combination. Replacing the gate by an additive branch of identical size and initial weights cannot be distinguished from removing it in either host (+0.4, +0.6, +0.1 pp and +0.8, +1.4, +9.3 pp; all intervals include zero), and the rescue ratios are small ($\rho = 0.029$, paired-seed interval $[0.011, 0.059]$; fully frozen host $\rho = 0.194$, $[-0.002, 0.326]$). Because gate removal is only weakened, the pre-specified rule labels the substitution test *not assessable*. This is a statement that no clear rescue was detected, not that the two arms are equivalent: in the fully frozen host the additive arm is +9.3 pp above the gateless arm at −6 dB (interval $[-7.2, 25.8]$, 5/5 seeds). The direct comparison is more informative: the multiplicative gate exceeds the additive replacement by +4.6, +7.9, +23.3 pp in the fully frozen host and by +9.0, +15.1, +26.8 pp in the partially frozen host, with intervals excluding zero at −2 and −6 dB in the former and at −6 dB in the latter (Table 4, descriptive). The ratio ρ is a conditional summary: among the 3125 paired seed resamples, at least one noise level was excluded because its denominator fell below 2 pp (as pre-specified) in 16.2% (507) of the resamples for the fully frozen host and in 0.5% (16) for the partially frozen host; no resample excluded all three levels (the repository field `paired_frac_dropped` is this second statistic, 0 for both).

4.2. Reverse transition under independent noise

A second pre-specified grid (60 cells, four arms; thresholds committed before the first cell) repeated the freeze and gate-removal contrasts on the reverse transition, 40 Hz/10 kN → 37.5 Hz/11 kN, with the same bearings in swapped roles, under independent noise. Relative to the earlier protocol the arm means moved by −1.8 to +1.7 pp. This direction is harder for the time-invariant model: on its evaluation set a majority-class predictor scores 46.8% macro-F1 and a uniformly random one 41.5%, and S4D sits close to those levels.

Table 5: Reverse transition (40 Hz/10 kN → 37.5 Hz/11 kN), independent training and evaluation noise. Macro-F1 (%), mean over 5 seeds, best-epoch / final-epoch; paired differences under final-epoch reporting with 95% t-intervals, seeds positive and label. Fully frozen gate removal and additive substitution were not run in this direction under independent noise (Table 3).

Arm (best / final)	0 dB	−2 dB	−6 dB	label
frozen−all	88.6 / 77.4	88.1 / 77.6	81.1 / 73.3	
BM3-frozen	85.5 / 75.4	86.7 / 78.4	78.5 / 72.2	
frozen−gate	64.0 / 56.1	62.5 / 55.7	55.1 / 50.1	
S4D	61.2 / 49.9	60.2 / 50.9	55.7 / 52.2	
frozen−all − S4D	+27.5 [21.3, 33.8] (5/5)	+26.7 [19.5, 33.9] (5/5)	+21.1 [16.2, 26.0] (5/5)	holds
BM3-frozen − S4D	+25.5 [13.9, 37.1] (5/5)	+27.5 [18.2, 36.9] (5/5)	+20.0 [8.0, 32.0] (5/5)	holds
BM3-frozen − frozen−gate	+19.3 [2.9, 35.7] (5/5)	+22.7 [4.1, 41.4] (5/5)	+22.0 [3.2, 40.9] (5/5)	holds
frozen−gate − S4D	+6.2 [−15.6, 28.0] (2/5)	+4.8 [−17.5, 27.1] (2/5)	−2.0 [−19.6, 15.5] (1/5)	descriptive

The fully frozen block exceeds S4D by +27.5, +26.7, +21.1 pp and the partially frozen block by +25.5, +27.5, +20.0 pp; removing the gate from the partially frozen block costs 19.3, 22.7, 22.0 pp, with intervals excluding zero and 5/5 seeds at every level. All three contrasts hold and none is reversed. In this direction gate removal is thus resolved at every level for the partially frozen host, whereas on the original transition it is resolved at −6 dB only; the two directions are reported side by side and not merged. The gateless arm cannot be distinguished from S4D (+6.2, +4.8, −2.0 pp). The contrasts with S4D rest on a baseline close to the trivial floor. The reverse direction shares the eight bearings of the forward one with swapped roles and has a

single inner-race bearing in its evaluation set, so it tests direction rather than new bearings.

4.3. Paderborn with shared bearings

On the Paderborn real-damage bearings (1500 → 900 rpm), a pre-specified 45-cell grid trained and evaluated on the same four bearings (KA04, KA16, KI04, KI14; different recordings) under independent noise (Table 6). Gate removal costs +7.6, +6.5, +4.4 pp with intervals excluding zero at all three levels (*holds*); the partially frozen block exceeds S4D by +5.6, +6.5, +8.3 pp and is resolved at −6 dB only (*weakened*). The fully frozen block, the additive arm and the graft were not run in this setting. The companion setting with bearings disjoint from the training set was run only under the earlier protocol, where per-index label agreement between training and evaluation windows is 0.999; it is reported in Appendix D and is not an independent-noise test. With two bearings per class a bearing's identity determines its label in the shared setting, so a model can succeed partly by recognising bearing-specific signatures (a form of shortcut learning [47]); XJTU-SY, by contrast, evaluates on bearings disjoint from the training ones in each direction.

Table 6: Independent training and evaluation noise, Paderborn shared bearings (KA04, KA16, KI04, KI14; 1500 → 900 rpm). Paired differences under final-epoch reporting with 95% t-intervals, seeds positive and pre-specified label. Means: BM3-frozen 81.5 / 83.8 / 85.9, S4D 75.9 / 77.3 / 77.6, frozen−gate 73.8 / 77.3 / 81.5 %.

Contrast	0 dB	−2 dB	−6 dB	label
BM3-frozen − S4D	+5.6 [−4.4, 15.5] (4/5)	+6.5 [−2.9, 15.9] (4/5)	+8.3 [2.7, 13.8] (5/5)	weakened
BM3-frozen − frozen−gate	+7.6 [2.7, 12.6] (5/5)	+6.5 [1.7, 11.3] (4/5)	+4.4 [3.2, 5.6] (5/5)	holds

4.4. Does the gate's contribution grow with noise depth?

A contrast that is resolved at one level and not at another does not by itself show that the effect differs between levels. We therefore computed, with seeds paired, the difference between the gate-removal cost at a deeper level and at 0 dB (Table 7). This analysis was added post hoc and is descriptive.

Only the fully frozen host on the forward transition shows a change whose interval excludes zero at −6 dB (+27.2 pp, 5/5 seeds). The partially frozen

Table 7: Change of the gate-removal cost (frozen host − its gateless counterpart, percentage points) relative to 0 dB: mean, 95% paired t-interval and seeds (of 5) with a positive change. Independent noise, final epoch. Descriptive; not pre-specified; the eight differences are not corrected for multiple comparisons.

Host and transition	−2 vs. 0 dB	−6 vs. 0 dB
Forward, fully frozen	+3.9 [0.5, 7.3] (4/5)	+27.2 [19.2, 35.2] (5/5)
Forward, partially frozen	+6.2 [−6.3, 18.7] (4/5)	+17.5 [−1.2, 36.2] (4/5)
Reverse, partially frozen	+3.4 [−1.6, 8.5] (4/5)	+2.7 [−15.7, 21.1] (2/5)
PU shared, partially frozen	−1.1 [−3.1, 0.9] (1/5)	−3.2 [−8.4, 1.9] (1/5)

host on the forward transition has a positive point estimate with an interval that includes zero; on the reverse transition the change is near zero although the cost itself is large at every level; for Paderborn shared bearings the point estimate is negative. The −2 versus 0 dB change for that block has a lower interval bound close to zero and is not treated as a separate finding. The statement that the gate's contribution grows with noise depth is therefore supported only by the fully frozen block on the forward transition, and we restrict it to that case throughout; for the other settings the data are compatible with a depth-independent cost.

4.5. Evidence from the earlier noise protocol

Several results were obtained only under the earlier protocol (Table 3, Appendix D). We list them with the weight we give them; none carries an abstract-level claim by itself.

- *Graft onto S4D* (Appendix D.4–Appendix D.5). A gate branch of the original width recovers $R = 0.196$ of the gap between S4D and BM3-frozen; one rebuilt in the donor's width recovers 0.296 (paired-seed interval $0.14, 0.72$), against 0.408 for an additive branch with identical parameters. On the original transition the gate did not recover more than the additive branch; on the reverse transition it recovered somewhat more ($\Delta R = 0.153$, paired-seed interval $0.01, 0.34$). Descriptive.
- *Additive substitution on the reverse transition* (Appendix D.6): not rescued by the pre-specified rule; not re-run under independent noise.

- *Stem and A removal* (Appendix D.3): no resolved effect of removing the heavy-tailed A parameterisation; removing the stem improves macro-F1 at 0 and -2 dB (intervals excluding zero) and costs 5.9 pp at -6 dB (interval including zero).

- *Per-class recall* (Appendix D.10, 20-epoch grid): gate-related differences concentrate on the minority inner-race class.

- *Paderborn with disjoint bearings* (Appendix D.8): none of the pre-specified contrasts replicates and no architecture transfers well (47.7–56.9%).

- *Reporting rule* (Appendix D.9): best-epoch values from the original grids are optimistic by 0.6–10.6 pp relative to the final epoch; orderings are unchanged.

5. Discussion

What can be credited? Within this Mamba-3 block, on this testbed and with noise-aware training, four statements are supported by independent-noise evidence. (i) Making the scan's dynamics and read/write coefficients input-independent did not remove the block's advantage over S4D on either transition of one dataset (the independent-noise grids contain no unfrozen BM3 arm, so they do not show that freezing preserves all of the unfrozen block's performance): S4D trails the frozen block by 12–28 pp. This does not show that input-dependence is useless: it shows that, in this block and under this budget, learned constants suffice to keep the advantage, and the two readings ("input-dependence contributes nothing here" and "what it contributes is recovered by constants") lead to the same attribution. (ii) The gate-removal cost is large in the partially frozen block (resolved at -6 dB on the original transition and at every level on the reverse one) and in the fully frozen block (resolved at -6 dB). (iii) Without its gate the fully frozen block still exceeds S4D at 0 and -2 dB, so the gate is not required for a positive advantage there; the removal cost at those levels (+5.4 and +9.3 pp) has intervals that include zero, so a contribution is neither shown nor excluded. At -6 dB the gateless block falls below S4D. (iv) A multiplicative gate exceeds an additive branch of identical size and initial weights by +23.3 pp at -6 dB in the fully frozen block and by +26.8 pp in the partially frozen block, whereas the additive arm cannot be distinguished from the gateless one; the

pre-specified rescue test is nevertheless not assessable and equivalence is not claimed.

Noise-dependent gating. Only one setting shows a gate contribution that grows significantly with noise depth: the fully frozen block on the original transition (change of the removal cost between 0 and $-6\,\mathrm{dB}$ $+27.2\,\mathrm{pp}$, interval excluding zero; Section 4.4). For the partially frozen block, the reverse transition and Paderborn shared bearings the data are compatible with a depth-independent cost, and the statement in the title is restricted to the first case. The depth analysis was added post hoc and is descriptive.

Relation to earlier work. Representability results for selectivity [16] are not contradicted by a single ablation on one testbed. The S4D-superiority finding [5] was obtained on clean classification benchmarks, without matched noise or a shift of operating condition, so it does not speak to the regime studied here. Two points of contact are nevertheless consistent with it: parity between S4D and the Mamba-style arms on clean data (Appendix D), and the finding that input-dependent selectivity is not needed for the block's advantage. What this paper adds is specific to the matched-noise, cross-condition regime studied here, not a contradiction of the clean-benchmark result.

Implications for backbone selection and for transplant practice. For practitioners choosing backbones for noisy, drifting signal regimes, the results suggest judging the block as a composition. Whether a gate added to a different host reproduces its effect was tested only under the earlier noise protocol (Appendix D.5): a gate built in the donor's shape recovered no more than an additive branch of the same size on the original transition and somewhat more on the reverse one, and neither design was tuned; we treat this as descriptive. A hypothesis for future work follows: a recomposed block guided by the component map (no conv stem, gate retained, S4D-style A) may match or exceed the original on this testbed; we leave it untested, because post-hoc recomposition after seeing the map would not carry the evidential status of the pre-specified chain.

Why might gating be composition-bound? We refrain from a mechanistic claim beyond the data, but note three compatible observations. First, the gate-related differences concentrate on the minority, lowest-recall class (Appendix D.10), and matched-noise training lifts that class *only when the gate is present*. One hypothesis, which we did not test, is that the gate is the component through which matched-noise training improves hard-class recall, i.e., that it exploits noise-aware training rather than acting as a test-time de-

noiser; we measured neither classification margins nor gate activations, and the reading is in any case bounded by the matched-SNR scope (Section 6). Second, the multiplicative form admits a per-sample suppression interpretation, but no gate statistics were recorded, so this remains a hypothesis rather than an observation. Third, the gateless composition falling to or below S4D at the deepest noise levels is compatible with the other components being calibrated to a gated regime (e.g., the stem may pass features that are only safe when the gate can veto them); this is the weakest of the three observations. The gateless–S4D comparison depends on host and protocol: in the fully frozen block under independent noise the gateless arm is above S4D at 0 and −2 dB and below it at −6 dB (Table 4), whereas for the partially frozen block under the earlier protocol it is not resolved against S4D with five seeds (Appendix D.9). All are testable in future work via pairwise interaction grids and mismatched-noise protocols.

6. Boundaries and Limitations

We state the scope plainly.

1. **Testbed and datasets.** *Testbed scope.* All claims are bounded to bearing vibration from one pair of operating conditions of one dataset (37.5 Hz/11 kN and 40 Hz/10 kN; speed and load both change), binary classification, GPU execution (no edge/real-time claims). Within each direction the training and evaluation bearings are disjoint; the reverse direction uses the same eight bearings with their roles swapped, has a single IR bearing in its evaluation set (the forward direction a single IR bearing in its training set) and so tests direction rather than new bearings; the third condition of the dataset has no IR bearing. The seeds do not resample bearings (Section 3.1), so the intervals carry no bearing-level uncertainty. Further independent datasets and multi-class settings are future work. *Second dataset.* On Paderborn, the gate-removal cost replicates when training and evaluation share bearings under independent noise, and the partially frozen block's advantage over S4D is resolved at −6 dB only. With bearings disjoint from the training set (earlier protocol only; per-index label agreement 0.999) nothing replicates and every architecture, S4D included, transferred poorly (47.7–56.9% against a balanced-chance level of about 50%). The two Paderborn settings differ in more than bearing overlap, so they do

not identify why transfer fails. The fully frozen block was run only in the disjoint setting. *Matched-SNR scope.* Training and evaluation share the same injected SNR throughout (Section 3.1); "robustness" in this paper is matched-noise robustness under noise-aware training, not clean-train/noisy-test generalisation. The gate's benefit may operate by exploiting matched-noise training (consistent with Appendix D.10) rather than by test-time denoising; attribution under mismatched-noise protocols is untested and may differ. *Noise-regime coverage.* Under strong coloured noise at −10 dB a TCN baseline slightly overtakes the Mamba-style block, and −10 dB is near-chance for the time-invariant baseline and excluded from adjudication; the demonstrated advantage regime is matched, Gaussian-dominated noise.

2. **Noise protocol and coverage.** *Noise protocol.* All grids not marked in Table 3 as independent-noise used training and evaluation noise shared by index. Where both protocols exist the arm means differ by at most 4 pp and no contrast changed direction, which is not a guarantee for the arms that were not re-run. *Coverage by protocol.* Table 3 lists, for every operation, whether it was run under independent noise; the graft onto S4D, the stem and A ablations, additive substitution on the reverse transition, the per-class grid and the disjoint-bearing Paderborn setting were not. Not run at all: gate removal and additive substitution in the fully frozen block on the reverse transition, and the fully frozen block, additive arm and graft on Paderborn shared bearings. *Transplant results are earlier-protocol and budget-conditional.* The graft onto S4D was run only under the earlier noise protocol, with a matched 50-epoch budget; a longer or graft-tuned budget might recover more, and we make no claim beyond *this gate branch, in this host, under a matched training budget* (Appendix D.4).
3. **What the interventions can and cannot show.** *Scope of the freezing intervention.* The first freeze arm removes the input-dependence of Δ, A, B, C only; the fully input-independent scan is a separate grid that was specified after the earlier grids had been seen. No equivalence margin was pre-specified, so "no measurable cost" means the point estimates favour the frozen arms with intervals including zero, not equivalence; the independent-noise grids contain no unfrozen BM3 arm, so they do not show that freezing preserves the unfrozen block's performance. The value stream and the gate input remain input-dependent, and freezing is implemented inside this Mamba-3 layer; a conclusion

about input-dependence in general would need other blocks and tasks. *Gate interventions.* Under independent noise, gate removal is resolved only at −6 dB in the partially and the fully frozen block on the original transition, and without its gate the fully frozen block still exceeds S4D at 0 and −2 dB; the combined statement that the fully frozen block's advantage is carried by the gate is therefore not made. Where the gate-removal cost is resolved at every level (partially frozen block, reverse transition) it was not tested in the fully frozen block. The additive substitution could not be distinguished from gate removal, but the multiplicative–additive differences are positive at −6 dB; with gate removal only weakened the pre-specified rescue rule is not assessable, and neither equivalence nor zero rescue is claimed. The rescue ratio is a conditional summary (levels with a denominator below 2 pp are excluded in some resamples). *Attribution level.* Credit assigned to "the rest of the composition" is by elimination within the enumerated component set; pairwise interaction terms are not decomposed. *Conv-stem dual reading.* Its clean-data cost admits both representation-mistransfer and optimisation-difficulty readings, reported in place (Appendix D.3). *Construction checks.* For the frozen arm the state-dictionary check compares names and shapes of non-target tensors; value-level comparison was applied to the component-ablation and graft arms only.

4. **Statistics, reporting and pre-specification.** *Sample size.* Every cell mean uses five seeds; run-to-run variance is a known confounder of benchmark conclusions [48, 49]. The smallest attainable one-sided Wilcoxon p is 0.031, paired intervals are wide, and several secondary statements (conv stem and A parameterisation; gateless versus S4D in the partially frozen block under the earlier protocol) are not resolved at this size. For the original-design grids conclusions rest on effect sizes that clear the pre-specified thresholds; for the later grids they rest on paired intervals as defined in Section 3.5. Seeds resample initialisation, batch order and noise (evaluation noise is one bank per seed), not bearings. *Reporting rule.* Reported values are final-epoch macro-F1 under a fixed 50-epoch budget. The best-epoch rule of the original grids, for which the early thresholds were pre-specified, is optimistic by 0.6–10.6 pp (Appendix D.9). The final-epoch re-run of the original-design arms (earlier noise protocol) and the independent-noise re-run cover different sets of cells (Table 3); levels outside the adjudication band and the per-class grid were not re-run. No source-domain validation rule

was available, and because the final epoch of the same evaluation condition has been inspected repeatedly, final-epoch values are not a fresh confirmatory test. *Pre-specification.* Thresholds were fixed before each grid: internally for the original design (file times, not tamper-evident) and, from the width-matched graft onwards, also in a public repository before the first cell (Section 3.5). The noise-depth analysis and the multiplicative–additive differences are post hoc.

A single-repeat check of process-level non-determinism ($\sim$0.7 pp) and the label-hash/adjudication trail are documented in the reproducibility appendices (Appendix A–Appendix B).

7. Conclusion

In this Mamba-3 block, on one controlled shift $\times$ noise testbed with matched-noise training and independent training and evaluation noise, making the scan's dynamics and read/write coefficients input-independent did not remove the block's advantage over a time-invariant S4D: the fully frozen block leads by +21.3, +23.3, +20.3 pp on the original transition and by +27.5, +26.7, +21.1 pp on the reverse one. Removing the multiplicative output gate causes a substantial loss at -6 dB, and this cost is resolved only at that level on the original transition (at every level for the partially frozen block on the reverse one; the ranking of the gate against the stem and A ablations comes from the earlier protocol and is descriptive), and without its gate the fully frozen block still exceeds S4D at 0 and -2 dB. Only in the fully frozen block on the original transition does the gate's contribution grow significantly with noise depth; in the other settings tested the data are compatible with a depth-independent cost. A multiplicative gate exceeds an additive branch of identical size at -6 dB in both hosts, although the pre-specified rescue test is not assessable. On Paderborn the gate-removal cost replicates when bearings are shared between training and evaluation. Grafting a gate onto S4D, per-class results, the stem and A ablations and the disjoint-bearing Paderborn setting were run only under the earlier noise protocol and are descriptive. The evidence rests on one block, one shift pair per dataset, five seeds and final-epoch reporting under a fixed budget. The independent-noise grids contain no unfrozen BM3 arm, so they establish that the frozen block beats S4D, not that freezing preserves the unfrozen block's performance. We offer the chain—freeze, remove, replace—as a protocol for

turning architecture comparisons into component-level tests, with two cautions from this study: interventions must be matched in width and capacity, and the noise streams of training and evaluation must be independent.

CRediT authorship contribution statement

Yi Wang: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization, Supervision, Project administration.

Yongqing Tang: Funding acquisition, Project administration, Writing – review & editing, Resources, Formal analysis.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used LLM-based agents to execute and cross-check pre-specified analysis and adjudication passes as described in Appendix C, and to assist with manuscript drafting and figure generation. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article. All numeric thresholds, adjudication rulings, and scientific interpretations reported in this paper were fixed and made by the human authors, as detailed in Appendix C.

Data availability

The code, pre-specification artifacts, and adjudication trail for this study are available at https://github.com/jeff-1980/bm3-jizi. The repository includes all follow-up grids (645 cells with per-epoch curves, including the 210 independent-noise cells of Sections 4.1–4.3), their pre-specifications, drivers, construction checks and decision memos and the harness's own imported modules. The analysis can be recomputed from the released per-cell records alone (it was re-run from a fresh clone and reproduced the archived decisions byte for byte). Retraining requires the XJTU-SY and Paderborn data, a GPU and the Mamba-3 source and kernels (upstream commit `e9594ce`); a portable loader resolves the harness's imports to the repository copies, and a 2-epoch smoke run of the reverse-transition driver from a clean copy of the repository (not the author's directory tree) loaded those copies and reproduced the evaluation-label hash, but a full retraining from a clean environment has not been performed.

Appendix A. Fairness protocol

Appendix A.1. Evaluation-set fingerprint

Every reported cell in this study is tagged with `eval_sha256`, a SHA-256 hash of the raw held-out test-label byte array, computed once per run and asserted identical across every arm/condition/seed cell within a grid. It fingerprints *that the same held-out label vector was scored every time*, not model weights or predictions. It is also the split on which the best-epoch rule of Section 3.3 selects epochs. The full value shared by all 865 cells across all five grids is:

```
6c20b367522ce5db59720120f8a7cd12792c7e070c6ae84bd1985511998727de
```

This value is recomputed independently at adjudication time (not merely copied from the run log) and cross-checked row-by-row against every grid's fairness report before any comparison across arms is treated as valid.

Appendix A.2. Per-arm parameter counts

Table A.8 lists every arm's parameter count, each obtained directly via `sum(p.numel() for p in model.parameters())` on the instantiated model (not estimated), cross-verified against an independently maintained architecture map.

Table A.8: Per-arm parameter counts for every arm of the study (from the `n_params` field recorded in each cell and the harness source).

Arm	Parameters
BM3 (`bm3_kin` / `bm3_nokin`)	177,938
BM3-frozen (`bm3_frozen`)	112,410
S4D (`s4d`)	100,162
S4D-wide (`s4d_wide`, capacity control, 1.75×)	178,498
frozen−conv (`frozen_noconv`)	112,026
frozen−gate (`frozen_nogate`)	79,642
frozen−A (`frozen_As4d`)	112,410
frozen−trap (`frozen_ctrap`)	111,906
frozen−rot (`frozen_cangle`)	104,346
frozen−all (`frozen_clti`)	103,842
frozen−all−gate (`frozen_clti_nogate`)	71,074
frozen−all+add (`frozen_clti_add`)	103,842
frozen+add (`bm3_frozen_add`)	112,410
S4D+gate (`s4d_plus_gate`)	116,802
S4D+gate, width-matched (`s4d_plus_gate_wm`)	198,978
S4D+additive branch (`s4d_plus_branch`)	198,978

Appendix A.3. Five-batch pre-specification timeline

Table A.9 lists, for each of the five pre-specified grids, the artifact that fixed its adjudication thresholds, that artifact's creation/modification date, and the corresponding grid's run-directory start time. In every case the prereg artifact predates the grid it governs.

Table A.9: Pre-specification predates data collection for all five grids.

Grid (cells)	Canonical prereg artifact	Prereg date	Grid start
Baseline (475)	Task-tracking-system card (`created: 2026-06-27`, mtime 15:02)	2026-06-27	2026-06-27 17:31
Frozen-selectivity (105)	`frozensel_prereg.json` (mtime 18:26:54)	2026-07-02	2026-07-02 18:27
Component ablation (175)	Task-tracking-system card (`created: 2026-07-03`), materialised in `prereg_provenance.md`	2026-07-03	2026-07-03 10:34
Graft (35)	`graft_prereg_provenance.md` (mtime 21:20:36)	2026-07-05	2026-07-05 22:41
Per-class (75)	Task-tracking-system card (`created: 2026-07-06`)	2026-07-06	2026-07-07 21:21

Two honest caveats. First, the baseline batch predates the orchestration infrastructure used for the other four batches by five days; its prereg trail rests on the task card's creation timestamp alone, not on the same automated executor/reviewer machinery documented in Appendix C, and we state this rather than implying a uniform process across all five batches. Second, for the component-ablation grid, two additional JSON files bearing "prereg" in their name exist on disk (`secondary_prereg_20260704-0708` and `secondary_deltapp_prereg_20260704-0719`) that *post-date* the grid's start time; both are explicitly marked void in the corresponding decision artifact's `void_preregs` field, and neither was used to adjudicate any result in this paper. They are the on-disk residue of the first post-hoc criterion substitution described in Appendix C, retained here (rather than deleted) as part of the released adjudication trail.

Appendix B. Confusion matrices and a repeat-run check

Appendix B.1. Full confusion-matrix set

Table B.10 gives the complete confusion-matrix set underlying the per-class analysis of Appendix D.10: 5 arms × 3 conditions (clean, 0 dB, −6 dB), each summed over 5 seeds (support OR= 17,520, IR= 8,720 per cell). These counts were dumped per-sample and re-verified against the reported recall/precision/F1 metrics cell-by-cell before use.

Table B.10: Full confusion-matrix set (summed over 5 seeds; support OR=17,520, IR=8,720 per cell). Columns give true→predicted counts.

Arm	Condition	OR→OR	OR→IR	IR→OR	IR→IR
BM3	clean	17,500	20	3,948	4,772
	0 dB	17,395	125	1,364	7,356
	−6 dB	16,613	907	1,664	7,056
BM3-frozen	clean	17,303	217	3,917	4,803
	0 dB	17,089	431	2,060	6,660
	−6 dB	15,634	1,886	2,543	6,177
S4D+gate	clean	17,327	193	4,969	3,751
	0 dB	16,303	1,217	3,322	5,398
	−6 dB	13,267	4,253	3,244	5,476
frozen−gate	clean	16,935	585	5,139	3,581
	0 dB	16,092	1,428	5,406	3,314
	−6 dB	13,200	4,320	5,469	3,251
S4D	clean	16,828	692	3,354	5,366
	0 dB	14,945	2,575	2,897	5,823
	−6 dB	12,477	5,043	3,810	4,910

Appendix B.2. Repeat-run check

Because two independent process runs of an identical configuration are not bitwise reproducible on this hardware (non-deterministic GPU kernel reductions in the selective-scan and normalisation ops), we measured a process-level noise floor directly: the same arm, condition, and seed (`bm3_frozen`, clean, seed 0, 2 epochs) was run twice in separate processes with identical evaluation fingerprints and identical parameter counts confirmed. The two runs' best macro-F1 differed by

$$|0.69025 - 0.69712| = 0.00687 \approx 0.687\,\text{pp},$$

which we report as an observed repeat difference of ~ 0.7 pp (cited in Section 3.5). It is a single repeat of one arm, one condition and a 2-epoch run, and therefore neither an estimate nor a bound of process variability for full-length runs. It is an order of magnitude below the seed-level variability observed within the deep-noise band across the curve tables underlying Figures D.2 and D.4 (standard deviations typically 2–9 pp, e.g. BM3-frozen 5.7–9.4 pp and S4D 1.7–4.5 pp at $\{0, -2, -6\}$ dB). Arm-level effects in Section 4 are judged against seed-level variability, not against this single repeat.

Appendix C. Automation disclosure and adjudication gates

Appendix C.1. Automation disclosure

The mechanical stages of the analysis pipeline—applying pre-specified numeric thresholds to aggregated results, re-computing aggregates, re-verifying evaluation fingerprints, and checking each decision against its pre-specified rule—were carried out by LLM-based coding agents under an orchestration harness, with every step auditable in the released trail. Manuscript drafting and figure generation were also assisted by such agents (main text, Declaration of Generative AI).

Appendix C.2. Incidents

Three incidents occurred in the original grids, each caught before an erroneous artefact entered a reported number: (1) on the frozen-selectivity grid an automated decision substituted an unspecified criterion for the pre-specified 10 pp threshold; re-applying the original rule gives the reported outcome, "not the primary mechanism"; (2) on the component-ablation grid an automated pass substituted an interval heuristic for the pre-specified threshold, and post-hoc rule files were generated; the true pre-specification predated the grid and was re-applied, and the post-hoc files are retained and marked void; (3) on the graft grid a construction stage was falsely marked as passed; the grid was re-executed from scratch under a corrected task card. Deviations in the later follow-up grids (for example a bootstrap made deterministic after first use, and rounding corrections) are recorded in the repository's decision memos. Full descriptions, with dates and artefact names, are given in Supplementary Material S1.

Appendix C.3. Human adjudication gates

All numeric adjudication thresholds, every ruling on an incident, the decision to report the incidents, and every interpretive claim in Section 5 were fixed or made by the human authors. No agent could overwrite a protected decision artefact, alter a pre-specified threshold or mark its own output as final.

Appendix C.4. Alignment with Elsevier's generative-AI policy

In line with Elsevier's policy, AI tools are not listed as authors and the human authors take full responsibility for the content. We extend the disclosure beyond the writing process to the adjudication pipeline because that is where automation could affect the reported conclusions.

Appendix D. Results under the earlier noise protocol

This appendix collects the results obtained while the noise added to window i of the training and the evaluation set was drawn from the same stream (Section 3.4), together with the final-epoch and best-epoch re-runs of the original grids. They are reported in full because they were the pre-specified adjudications of the original design, but they are of lower evidential weight than the independent-noise results of Section 4: no abstract-level claim rests on them alone, thresholds labelled "primary" or "necessary" below refer to the earlier-protocol grid only, and where an independent-noise counterpart exists (Table 3) the counterpart takes precedence. Section 4.5 summarises what these results add.

Appendix D.1. Freezing the input-dependent Δ, A, B, C projections (earlier protocol)

Freezing the input-dependent Δ, A, B, C projections (Section 3.2) costs at most 3.7 pp anywhere in the adjudication band on the point estimates under final-epoch reporting ($+3.7$, $+0.9$, $+0.2$ pp at 0, -2, -6 dB), while the frozen block retains an advantage of $+15.5$, $+19.8$, $+16.4$ pp over S4D (Table D.11, Figure D.2). Five seeds bound these quantities loosely: paired 95% intervals for Δ(BM3$-$frozen) are $[-3.8, 11.3]$, $[-4.5, 6.3]$, $[-12.1, 12.6]$ pp (BM3 ahead in $4/5$, $3/5$, $2/5$ seeds), and for Δ(frozen$-$S4D) $[4.3, 26.8]$, $[12.0, 27.7]$, $[4.1, 28.7]$ pp ($5/5$, $5/5$, $4/5$ seeds). The pre-specified rule attributed robustness to selectivity only if Δ(BM3$-$frozen) ≥ 10 pp in deep noise with the frozen arm collapsing toward S4D. It was written for best-epoch reporting,

Table D.11: BM3, BM3-frozen and S4D, macro-F1 (%), mean over 5 seeds; Δ columns are seed-paired mean differences in percentage points. Upper block: final epoch of the 50-epoch budget on the adjudication band (primary; re-runs of Appendix D.2 and Appendix D.9). Middle block: best epoch on the evaluation condition in the same re-runs (sensitivity). Lower block: best epoch in the original frozen-selectivity grid for levels outside the band (auxiliary; not re-run). Earlier noise protocol (training and evaluation noise shared by index); see Section 4.1 for the independent-noise re-run.

Condition	BM3	BM3-frozen	S4D	Δ(BM3−frozen)	Δ(frozen−S4D)
Final epoch (primary)					
0 dB	86.9	83.1	67.6	+3.7	**+15.5**
−2 dB	88.3	87.3	67.5	+0.9	**+19.8**
−6 dB	83.3	83.1	66.7	+0.2	**+16.4**
Best epoch (sensitivity, same re-runs)					
0 dB	95.6	90.6	75.9	+5.1	+14.6
−2 dB	95.8	92.3	75.1	+3.5	+17.2
−6 dB	88.6	85.7	68.6	+3.0	+17.1
Auxiliary (best epoch, original grid)					
clean	78.9	78.9	81.0	−0.0	−2.1
−10 dB*	68.1	64.3	52.3	+3.8	+12.0

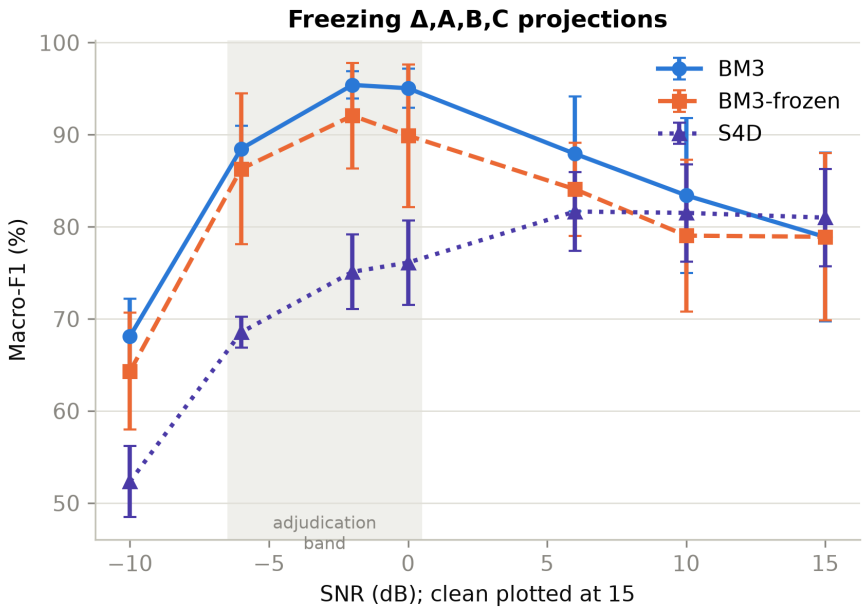


Figure D.2: Macro-F1 vs. SNR for BM3, BM3-frozen, and S4D across the full noise grid (best epoch, original frozen-selectivity grid; final-epoch values on the adjudication band are in Table D.11, and levels outside the band are auxiliary). The frozen arm tracks BM3 closely and stays well above S4D throughout the adjudication band. Earlier noise protocol (training and evaluation noise shared by index).

under which the original grid gives Δ(BM3−frozen) = +5.1, +3.3, +2.2 pp (re-run: +5.1, +3.5, +3.0) and Δ(frozen−S4D) = +13.8, +17.0, +17.7 pp. Neither condition holds under either rule, so the pre-specified verdict is that these projections are *not the primary mechanism*. This is a failure to meet a confirmation criterion, not a demonstration that freezing is cheap: the intervals above are compatible with differences larger than 10 pp, and no equivalence margin was pre-specified or tested. One limit applies: the frozen arm still contains input-dependent trapezoidal weights and rotation angles, so this result concerns Δ, A, B, C only; Appendix D.2 removes that limit. On clean data (auxiliary, best epoch) S4D matches or slightly exceeds both Mamba-style arms—in line with the clean-benchmark comparison of [5], which, like this auxiliary row, involves no added noise. In the noisy regime, the block's advantage over S4D is not removed by freezing these four terms, so this freeze does not identify where the advantage comes from; the other parts of the block and the remaining input-dependent scan terms are candidates.

Appendix D.2. Freezing the remaining input-dependent scan terms (earlier protocol)

Table D.12 reports the three layered arms of Section 3.2 on the adjudication band (60 cells: 4 arms × 3 levels × 5 seeds × 50 epochs, the original training code, the same evaluation fingerprint, with the original BM3 arm re-run in the same grid). The question is whether the advantage that survived freezing Δ, A, B, C is carried by the two input-dependent terms the frozen arm retained.

It is not. With the scan's dynamics and read/write coefficients input-independent, *frozen−all* is +21.9, +24.0 and +21.2 pp above S4D at 0, −2 and −6 dB under final-epoch reporting, positive in 5/5 seeds at every level and with intervals excluding zero—if anything a larger margin than the partially frozen arm's +15.5, +19.8, +16.4 pp (Table D.20). None of the three freezing steps shows an accuracy loss in the point estimates: the paired differences BM3-frozen − frozen−trap, − frozen−rot and − frozen−all are negative at all three levels (between −1.8 and −6.3 pp under final-epoch reporting), i.e. the more heavily frozen arm is numerically ahead, with every interval including zero. The direction is not an artefact of capacity: *frozen−all* has 103,842 parameters against 112,410 for BM3-frozen and 177,938 for BM3. At the deepest level *frozen−all* also exceeds the full BM3 block by 4.5 pp ([−7.2, −1.9] for BM3 − frozen−all, 0/5 seeds in BM3's favour); we

Table D.12: Layered freezing of the scan's input-dependent terms. Macro-F1 (%), mean over 5 seeds, under best-epoch and final-epoch reporting; Δ columns are paired differences under final-epoch reporting with 95% t-intervals over seeds. Earlier noise protocol (training and evaluation noise shared by index); see Section 4.1 for the independent-noise re-run.

	0 dB		−2 dB		−6 dB	
Arm	best	final	best	final	best	final
BM3 (re-run)	95.6	86.9	95.8	88.3	88.6	83.3
BM3-frozen (Δ, A, B, C)	90.6	83.1	92.3	87.3	85.7	83.1
frozen−trap	95.5	87.6	95.3	89.3	90.5	88.0
frozen−rot	94.6	86.7	94.6	89.1	89.0	86.4
frozen−all	96.5	89.5	96.8	91.5	90.5	87.9
S4D	75.9	67.6	75.1	67.5	68.6	66.7
Paired differences, final-epoch reporting (pp)						
frozen−all − S4D	+21.9 [16.7, 27.1]		+24.0 [21.4, 26.5]		+21.2 [17.5, 24.9]	
BM3-frozen − frozen−all	−6.3 [−17.6, 4.9]		−4.1 [−12.0, 3.7]		−4.8 [−17.8, 8.2]	
BM3 − frozen−all	−2.6 [−8.2, 2.9]		−3.2 [−7.0, 0.6]		−4.5 [−7.2, −1.9]	

report this but do not build on it, since five seeds and one speed pair cannot separate a genuine advantage of a time-invariant scan from a variance effect.

Two readings of the whole step are therefore available, and the data do not separate them: input-dependence of the scan contributes nothing measurable here, or it contributes something that the learned constants recover under this training budget. Either way the conclusion for the attribution chain is the same and now rests on a complete intervention rather than a partial one: **the Mamba-style block's deep-noise advantage over a time-invariant S4D does not require any input-dependent term in its scan**. What remains to carry it is the rest of the block—the convolutional stem, the heavy-tailed A parameterisation, the normalisations and the output gate—which is what Appendix D.3–Appendix D.4 examine. The re-run also reproduces the original grids: the BM3 arm's best-epoch means (95.6, 95.8, 88.6) exceed the original frozen-selectivity grid's 95.0, 95.4, 88.5 by 0.57, 0.43 and 0.16 pp (a different PyTorch/Triton build, Appendix D.9).

Appendix D.3. Single-component removal on the frozen base (earlier protocol)

"Necessary" is used here in the operational sense of this ablation: removing the gate from the frozen block, with the other components and the budget unchanged, removes most of the block's advantage. The gate result is large,

Table D.13: Single-component removal on the frozen base, Δ(frozen − arm) in percentage points: final-epoch mean, 95% t-interval over seeds and number of seeds (of 5) with a positive difference (re-runs of Appendix D.9); "best" is the best-epoch value of the original component-ablation grid, for which the thresholds were pre-specified. Earlier noise protocol (training and evaluation noise shared by index); see Section 4.1 for the independent-noise re-run.

Component removed	0 dB	−2 dB	−6 dB	Adjudication
SiLU gate	+9.1 [2.9, 15.3] (5/5); best +13.1	+15.6 [−0.8, 32.0] (4/5); best +21.1	+22.6 [5.8, 39.4] (5/5); best +23.5	PRIMARY (all $\geq$ 8 pp, both rules)
conv stem (k=7)	−13.2 [−24.1, −2.2] (0/5); best −6.6	−7.5 [−14.6, −0.3] (0/5); best −2.7	+5.9 [−4.0, 15.8] (3/5); best +7.9	outside bins; regime trade-off
heavy-tailed A → S4D-style A	−1.9 [−9.2, 5.3] (2/5); best −2.4	−2.6 [−8.6, 3.4] (2/5); best −2.2	−4.8 [−16.4, 6.9] (2/5); best −4.2	outside bins; intervals include 0

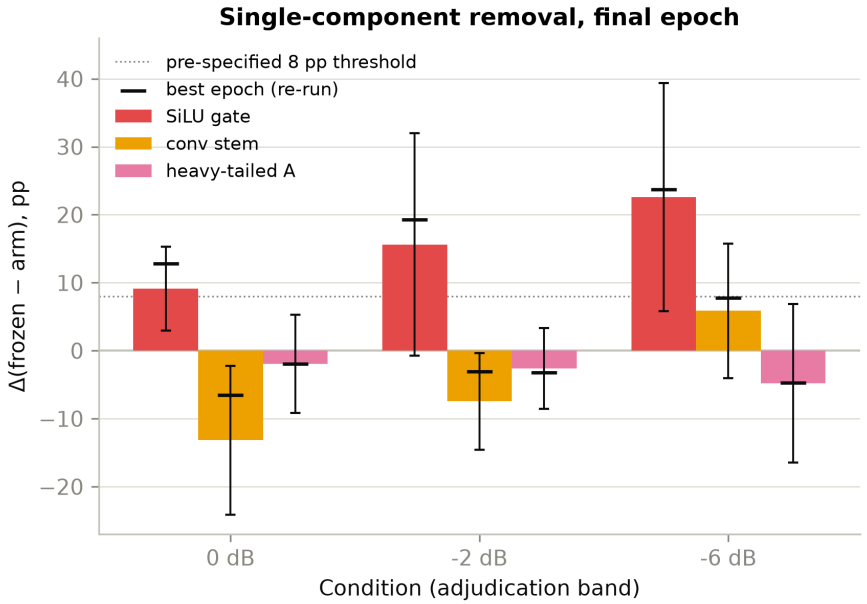


Figure D.3: Component-removal Δ (percentage points, frozen − arm) on the adjudication band for the three single-component ablations on the frozen base: final-epoch means with 95% t-intervals over seeds (bars) and best-epoch means of the same re-runs (black ticks). Only gate removal is large and one-signed at every level and clears the pre-specified 8 pp threshold. Earlier noise protocol (training and evaluation noise shared by index).

one-signed and grows with noise depth: under final-epoch reporting removal costs 9.1, 15.6, 22.6 pp at 0, −2 and −6 dB (paired 95% intervals [2.9, 15.3], [−0.8, 32.0], [5.8, 39.4]; 5/5, 4/5, 5/5 seeds), above the pre-specified 8 pp threshold at every level; the interval at −2 dB includes zero because of the large seed variance of the gateless arm. Under the best-epoch rule for which the threshold was written, the component-ablation grid gives 13.1, 21.1 and 23.5 pp ([6.5, 19.6], [3.8, 38.5], [8.5, 38.6]; 5/5 seeds at every level), so under the earlier noise protocol the verdict is *primary* under both rules. Under independent training and evaluation noise the same contrast is resolved only at −6 dB (Section 4.1), so the operational sense of "necessary" holds there at the deepest level only. The gateless arm gives up most of the Mamba advantage but is not resolved against plain S4D: frozen−gate − S4D is +6.4, +4.2, −6.2 pp, with every interval including zero. We therefore describe it as roughly at the level of S4D, below it only on the point estimate at −6 dB; this suggests, but does not establish, that components interact rather than add. The other two components tell regime stories rather than robustness stories. Removing the convolutional stem is beneficial at 0 and −2 dB (frozen − frozen−conv = −13.2 [−24.1, −2.2] and −7.5 [−14.6, −0.3], with the stemless arm ahead in every seed) and costly at −6 dB (+5.9, [−4.0, 15.8], 3/5 seeds); under best-epoch reporting the pattern was the same (−6.6, −2.7, +7.9 pp). On clean cross-speed data (auxiliary, best epoch, original grid) the stem costs 21.0 pp (removal yields 98.8 vs. 77.8, with seed-std collapsing from 8.1 to 0.6 pp)—consistent either with speed-specific local temporal patterns mistransferring across conditions, or with optimisation friction under the matched budget; we report both readings. Replacing the heavy-tailed state parameterisation by an S4D-style one is numerically beneficial at all three levels (by 1.9, 2.6, 4.8 pp under final-epoch and 2.4, 2.2, 4.2 pp under best-epoch reporting), but every paired interval includes zero, so we draw no directional conclusion from it.

Appendix D.4. Adding a narrow gate branch to S4D (earlier protocol)

The pre-specified recovery ratio is R = mean over the band of (S4D+gate− S4D)/(frozen − S4D). Under final-epoch reporting R = **0.196** (per level 0.249, 0.341, −0.003; seed-resampled 95% interval [0.09, 0.31]); under the best-epoch rule for which it was specified, $R = 0.174$ ([0.04, 0.29]). The pre-specified bins were $R \geq 0.7$ (confirmed), $0.3 \leq R < 0.7$ (partial recovery) and $R < 0.3$ (insufficient); both values fall in the last, although the final-epoch interval reaches the 0.3 boundary. The per-level paired gains of S4D+gate over

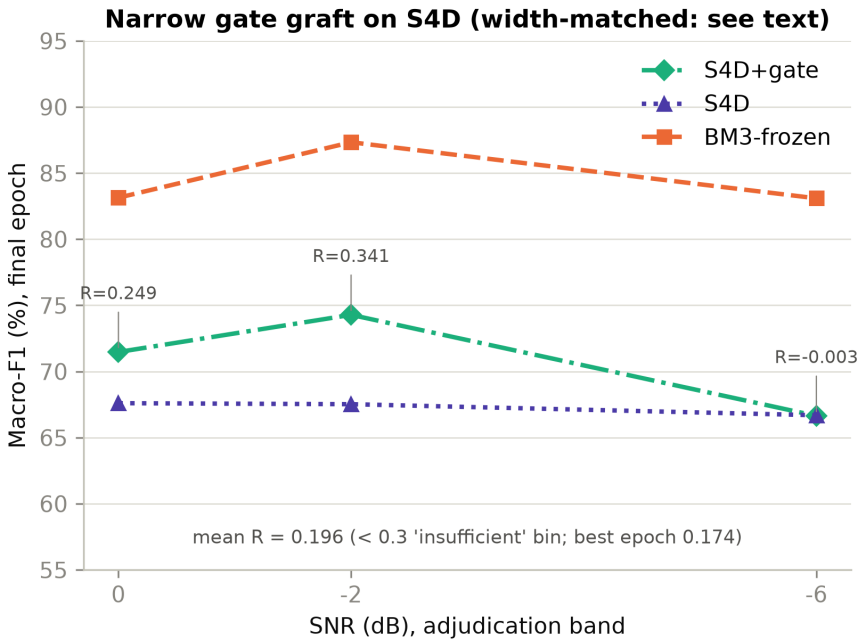


Figure D.4: Graft outcome under final-epoch reporting: S4D+gate, plain S4D, and BM3-frozen on the adjudication band (one re-run grid), with the per-level recovery ratio $R = (\text{S4D+gate} - \text{S4D})/(\text{frozen} - \text{S4D})$ annotated. Recovery is near zero at the deepest adjudicated noise level. This is the narrow graft; the width-matched graft and its additive control are in Table D.15. Earlier noise protocol (training and evaluation noise shared by index).

Table D.14: Grafting the SiLU gate branch onto plain S4D (S4D unchanged key-by-key; only the gate branch added). Upper block: final epoch, all three arms from the same re-run (Appendix D.9). Lower block: best epoch as pre-specified, with S4D+gate from the graft grid and S4D and BM3-frozen, and hence R, from the component-ablation grid of Appendix D.3 (independent runs of the same arms differ by 0.1–0.7 pp). Earlier noise protocol (training and evaluation noise shared by index); see Section 4.1 for the independent-noise re-run.

Condition	S4D+gate	S4D	BM3-frozen	Recovery ratio
Final epoch (primary)				
0 dB	71.5	67.6	83.1	0.249
−2 dB	74.3	67.5	87.3	0.341
−6 dB	66.6	66.7	83.1	−0.003
Best epoch (as pre-specified)				
0 dB	79.8	76.2	90.6	0.249
−2 dB	79.4	75.2	92.8	0.239
−6 dB	69.2	68.6	86.0	**0.035**

S4D under final-epoch reporting are +3.9 ([−0.8, 8.6]), +6.8 ([2.4, 11.1]), −0.1 ([−4.0, 3.9]) pp. At the deepest adjudicated level the graft recovers nothing. Removal and addition are not mirror-image interventions. In BM3 the gate has its own 64 → 128 projection and multiplies the scan output at the inner width of 128 before the output projection; removing it deletes 32,768 parameters. In the S4D host the grafted gate is a 64 → 64 linear branch (with bias) computed from the layer input and multiplying the SSM output before the residual addition; it adds 16,640 parameters over four layers. The two share the multiplicative SiLU form but not width, projection structure or parameter count. Removal costs 9–23 pp in the frozen block, whereas this graft changes S4D by between −0.1 and +6.8 pp (final epoch). What this shows is that *this gate branch, in this host and under this budget, recovers little*. Three readings fit the data and cannot be separated here: (i) host dependence, i.e. the gate needs the stem or scan context it co-trained with; (ii) implementation differences in width, projection and placement; (iii) transplant friction, i.e. the grafted branch is harder to optimise in a foreign host under the matched budget (training converges normally, but convergence is not robustness). Appendix D.5 tests reading (ii) with a width-matched graft and an equal-capacity non-multiplicative control; a tuning or longer-budget

comparison selected on a source-domain validation set was not run.

The four arms do allow a coarse host-by-gate contrast, $I = [F(\text{frozen}) - F(\text{frozen}{-}\text{gate})] - [F(\text{S4D}{+}\text{gate}) - F(\text{S4D})]$, where F is macro-F1. Under last-epoch reporting $I = 5.2$ [2.3, 8.1], 8.9 [−6.2, 23.9] and 22.6 [2.9, 42.4] pp at 0, −2 and −6 dB (seed-matched 95% t-intervals; positive in 5/5, 4/5 and 5/5 seeds); under best-epoch reporting in the re-run it is 7.8 [1.4, 14.2], 14.4 [−1.4, 30.3] and 23.1 [8.7, 37.4] pp. A positive I means the gate adds more on the frozen host than on S4D. Because the host differs in several respects at once and the two gates are not aligned, I cannot be assigned to a specific component or interaction term, and the seed pairing across arms is nominal (arms share seed indices, not initialisations).

Appendix D.5. A width-matched graft and an additive control (earlier protocol)

The graft of Appendix D.4 is narrower and smaller than the donor gate. Two further arms rebuild it in the donor's shape (Table 1). In each layer the layer input is projected to x and z of width 128 (the frozen block's inner width), the S4D layer runs on x at width 128 with state size 64—so that its own parameter count and total state size equal the host's 64-wide, state-128 layer—and an output projection maps back to width 64. *S4D+gate (wm)* combines $y \odot \text{SiLU}(z)$, as the donor does; *S4D+branch* uses the same modules, parameters and initial weights but combines $y + \text{SiLU}(z)$. Both have 198,978 parameters, more than BM3-frozen (112,410) and BM3 (177,938), which biases the test towards recovery. A construction check with forward hooks confirms that the tensor entering the output projection equals the intended formula for each arm and not the other, and that the two arms are identical at initialisation. The thresholds were committed publicly before the grid ran (Section 3.5): R as before, under final-epoch reporting; the verdict is INCONCLUSIVE whenever the seed-resampled interval crosses a bin boundary, in which case we write at the tier less favourable to the claim; and a gate-specific reading requires $\Delta R = R_{\text{gate}} - R_{\text{branch}} > 0.1$ with an interval excluding zero.

The width-matched graft recovers $R = 0.296$ of the gap (per level 0.159, 0.298, 0.431). Its interval, [0.16, 0.52] under the pre-specified bootstrap ([0.14, 0.72] under paired-seed resampling), crosses the 0.3 boundary, so the pre-specified verdict is INCONCLUSIVE between "insufficient" and "partial", and we report it at the less favourable tier: a graft built in the donor's shape recovers part of the gap. The recovery is not gate-specific. The additive

Table D.15: Width-matched graft and additive control on the original transition. Macro-F1 (%), mean over 5 seeds, final-epoch reporting; R is the mean over the three levels of (arm − S4D)/(BM3-frozen − S4D), with a seed-resampled 95% interval. S4D, BM3-frozen and the narrow graft are the re-runs of Appendix D.9. Earlier noise protocol (training and evaluation noise shared by index); see Section 4.1 for the independent-noise re-run.

Arm	0 dB	−2 dB	−6 dB	R	95% interval
S4D	67.6	67.5	66.7	–	–
S4D+gate (narrow)	71.5	74.3	66.6	0.196	[0.09, 0.31]
S4D+gate (wm)	70.1	73.4	73.8	0.296	[0.16, 0.52]
S4D+branch	73.1	76.2	73.8	0.408	[0.31, 0.60]
BM3-frozen	83.1	87.3	83.1	1	–
ΔR = S4D+gate (wm) − S4D+branch				−0.112	[−0.26, 0.02]

branch, with the same parameters, recovers at least as much ($R = 0.408$), $\Delta R = -0.112$ $[-0.26, 0.02]$ (paired-seed $[-0.20, 0.06]$), and at 0 dB the gated arm is below the additive one by 3.0 pp ($[-4.5, -1.5]$, 0/5 seeds). With frozen−all in the denominator the gated arm's ratio is 0.231 $[0.13, 0.33]$; under best-epoch reporting the two ratios are 0.459 and 0.483. What a graft recovers on this transition therefore follows the added structure—a wider channel axis and input/output projections—rather than the multiplicative interaction. Reading (ii) of Appendix D.4 is partly confirmed, in that the narrow graft understated what added structure can recover; but in the S4D host the gate did not contribute measurably beyond an equal-capacity additive branch, whereas removing it from the frozen Mamba-3 block costs 9–23 pp under the same rule (Table D.20).

Appendix D.6. Replacing the gate in its native host (earlier protocol)

The previous subsection changes the host and keeps the operator. A pre-specified grid (15 cells, original transition) does the converse: it keeps the host and changes the operator. In *frozen+add* the scan kernel of every BM3-frozen layer is called without gating and the layer output becomes $W_{\text{out}}(y+\text{SiLU}(z))$; modules, parameter count (112,410) and initial weights are bitwise those of BM3-frozen, and a construction check confirms that the donor operator is $y \odot \text{SiLU}(z)$ and that the new arm computes $y + \text{SiLU}(z)$. The pre-specified statistic is the native rescue ratio $\rho = (\text{add} - \text{no gate})/(\text{frozen} - \text{no gate})$, mean over levels, final epoch, with a seed-resampled interval; *not rescued* requires $\rho < 0.3$ with an upper bound below 0.5.

Table D.16: Gate replaced by an additive branch inside BM3-frozen, original and reverse transitions. Macro-F1 (%), mean over 5 seeds; paired differences under final-epoch reporting with 95% t-intervals and the number of seeds (of 5) with a positive difference. Earlier noise protocol (training and evaluation noise shared by index); see Section 4.1 for the independent-noise re-run.

	0 dB		−2 dB		−6 dB	
Arm	best	final	best	final	best	final
Original transition (37.5 Hz/11 kN → 40 Hz/10 kN)						
BM3-frozen ($y \odot \mathrm{SiLU}(z)$)	90.6	83.1	92.3	87.3	85.7	83.1
frozen+add ($y + \mathrm{SiLU}(z)$)	78.0	74.7	72.6	71.5	62.3	60.9
frozen−gate (no z)	77.7	74.0	73.1	71.7	61.9	60.5
BM3-frozen − frozen+add	+8.5 [1.1, 15.9] (5/5)		+15.8 [−1.2, 32.9] (4/5)		+22.2 [4.6, 39.9] (5/5)	
frozen+add − frozen−gate	+0.6 [−1.7, 3.0] (3/5)		−0.2 [−3.4, 3.0] (3/5)		+0.4 [−0.7, 1.4] (3/5)	
Reverse transition (40 Hz/10 kN → 37.5 Hz/11 kN)						
BM3-frozen ($y \odot \mathrm{SiLU}(z)$)	81.1	74.7	84.0	76.7	80.8	73.9
frozen+add ($y + \mathrm{SiLU}(z)$)	64.3	56.5	62.2	54.4	55.7	50.5
frozen−gate (no z)	63.8	56.2	62.0	54.7	57.1	50.0
BM3-frozen − frozen+add	+18.1 [1.2, 35.0] (5/5)		+22.3 [3.5, 41.0] (5/5)		+23.4 [6.8, 40.0] (5/5)	
frozen+add − frozen−gate	+0.3 [−0.6, 1.2] (3/5)		−0.3 [−1.0, 0.4] (2/5)		+0.6 [−0.2, 1.4] (5/5)	

The additive branch does not measurably rescue the loss. frozen+add cannot be distinguished from the gateless arm at any level (+0.6, −0.2, +0.4 pp, every interval including zero), and the multiplicative arm exceeds it by 8.5, 15.8, 22.2 pp. The rescue ratio is $\rho = 0.024$ (per level 0.070, −0.015, 0.016), with a 95% interval of $[-1.159, 0.486]$ under the pre-specified bootstrap, which resampled each arm and level independently; resampling seeds jointly across arms and levels, as the paired design warrants, gives $[-0.043, 0.136]$. Under best-epoch reporting $\rho = 0.005$. By the pre-specified rule the verdict is *not rescued*: the point-estimate rescue is small and below the pre-specified criterion, which is not a demonstration that the two arms are equivalent.

A further pre-specified grid (15 cells) repeats the substitution on the reverse transition (Appendix D.7), with BM3-frozen and frozen−gate taken from the reverse grid. The outcome is the same (Table D.16, lower block): frozen+add cannot be distinguished from the gateless arm at any level (+0.3, −0.3, +0.6 pp, every interval including zero), and the multiplicative arm exceeds it by +18.1, +22.3, +23.4 pp (intervals excluding zero, 5/5 seeds at every level). The rescue ratio is $\rho = 0.009$ (per level 0.017, −0.014, 0.023), with a 95% interval of $[-0.943, 0.362]$ under the pre-specified bootstrap and $[0.002, 0.018]$ under paired-seed resampling; under best-epoch reporting $\rho = -0.005$. The pre-specified verdict is again *not rescued*.

Together with Appendix D.5 this gives the two halves of a symmetric test. In the S4D host the gate did not recover more than an additive branch of the same size on the original transition and recovered somewhat more on the reverse one; in the Mamba-3 host an additive branch of the same size could not be distinguished from removing the gate, on both transitions. Under the earlier noise protocol this pattern is specific to the multiplicative operator in its native host; Section 4.1 shows that the size of the gate's contribution there depends on noise depth.

Appendix D.7. The reverse transition (earlier protocol)

A pre-specified grid repeats the chain's central contrasts on the reverse transition, 40 Hz/10 kN → 37.5 Hz/11 kN (75 cells: frozen−all, BM3-frozen, frozen−gate, S4D and S4D+gate (wm), at three levels and five seeds; Section 3.1 for the data). The independent-noise re-run of four of these arms is Section 4.2. This direction is harder for the time-invariant model: on its evaluation set, predicting the majority class gives a macro-F1 of 46.8%

and uniform random guessing 41.5%, and S4D sits close to those levels (Table D.17).

Table D.17: Reverse transition (40 Hz/10 kN → 37.5 Hz/11 kN). Macro-F1 (%), mean over 5 seeds, best-epoch and final-epoch reporting; paired differences under final-epoch reporting with 95% t-intervals and the number of seeds (of 5) with a positive difference. Earlier noise protocol (training and evaluation noise shared by index); see Section 4.1 for the independent-noise re-run.

Arm (best / final)	0 dB	−2 dB	−6 dB
frozen−all	86.3 / 77.5	87.8 / 77.9	81.7 / 74.0
BM3-frozen	81.1 / 74.7	84.0 / 76.7	80.8 / 73.9
S4D+gate (wm)	79.0 / 72.9	80.5 / 73.9	78.5 / 72.2
frozen−gate	63.8 / 56.2	62.0 / 54.7	57.1 / 50.0
S4D	61.2 / 51.2	61.0 / 52.4	54.6 / 51.8
frozen−all − S4D	+26.3 [21.5, 31.0] (5/5)	+25.6 [21.1, 30.1] (5/5)	+22.2 [18.8, 25.6] (5/5)
BM3-frozen − frozen−gate	+18.4 [1.2, 35.6] (5/5)	+22.0 [3.4, 40.6] (5/5)	+24.0 [7.6, 40.3] (5/5)
frozen−gate − S4D	+5.0 [−17.6, 27.7] (2/5)	+2.4 [−19.2, 23.9] (2/5)	−1.8 [−18.9, 15.3] (1/5)
S4D+gate (wm) − S4D	+21.7 [15.4, 28.0] (5/5)	+21.5 [17.0, 26.0] (5/5)	+20.5 [15.7, 25.2] (5/5)

The first two steps replicate. The fully input-independent scan exceeds S4D by +26.3, +25.6 and +22.2 pp at 0, −2 and −6 dB, with intervals excluding zero and 5/5 seeds at every level, so by the pre-specified rule (at least two of three levels) the first step is replicated on the reverse transition. Removing the gate from BM3-frozen costs 18.4, 22.0 and 24.0 pp, above the 8 pp threshold at all three levels, with intervals excluding zero and 5/5 seeds, and the gateless arm is indistinguishable from S4D (intervals include zero). On the third step, which the pre-specification treats as descriptive here, the width-matched graft recovers most of the gap, $R = 0.911$ $[0.82, 1.04]$ (0.863 with frozen−all in the denominator). A further pre-specified grid (15 cells) adds the same-size additive control in this direction: it reaches 67.8, 72.1, 68.4% (final epoch) and recovers $R = 0.758$ $[0.66, 0.88]$. The gate arm exceeds it by +5.1, +1.7, +3.8 pp (only the −6 dB interval excludes zero), and the difference in recovery is $\Delta R = 0.153$ $[0.06, 0.26]$; under best-epoch reporting it shrinks to 0.037 $[-0.1, 0.17]$. By the pre-specified descriptive labels this is *unresolved*: in this direction the gate carries a small increment

over the additive branch, whereas on the original transition it carried none ($\Delta R = -0.112$). Most of the reverse-direction recovery is therefore available to an additive branch of the same size, and the remaining gate increment is direction-dependent. Both ratios rest on an S4D baseline close to the trivial floor (51–52% against 46.8% for the majority class). Both statements carry the limits of this pair: the same eight bearings with their roles swapped, and a single inner-race bearing in the evaluation set.

Appendix D.8. Paderborn: disjoint and shared bearings (earlier protocol)

A pre-specified grid (75 cells; thresholds, a ceiling-and-floor check and the writing consequences committed before the first cell) repeats the three central contrasts on the Paderborn KAt data [50], using real-damage bearings only. Models train on two outer-race (KA04, KA16) and two inner-race bearings (KI04, KI14) at 1500 rpm and are evaluated on five different bearings (KA15, KA22; KI17, KI18, KI21) at 900 rpm, with load and radial force unchanged; the vibration channel is resampled to 25.6 kHz so that a window spans 80 ms as in XJTU-SY, and everything else (noise, training, arms, budget) is as in the main grids. The pre-specified smoke check (two arms, one extra seed) passed its three criteria, the learnability criterion by 0.3 pp. A second pre-specified grid (45 cells, three arms) then used the same four bearings for training and evaluation, at the same two speeds, from different recordings. This changes the evaluation bearings, their damage profiles, the number of windows and the class balance together, so the two settings are two protocols rather than a single-factor manipulation.

None of the pre-specified contrasts replicates (Table D.18). The fully input-independent block does not exceed S4D (frozen−all − S4D: −1.4, −2.5, −3.3 pp, the interval at −6 dB excluding zero in the negative direction), nor does BM3-frozen (−3.8, −3.6, −3.4 pp). Removing the gate shows no accuracy cost in this setting: BM3-frozen − frozen−gate is −5.0, −4.6, −3.3 pp, below zero in every seed at every level. With no gate effect to rescue, the native-substitution test is not assessable; the additive and gateless arms again cannot be distinguished (0.0, 0.0, −0.1 pp). The pre-specified labels are therefore *not replicated* for both freeze contrasts and for gate removal, and *not assessable* for the substitution test.

Two observations bound how this result can be read, and neither changes the labels. First, the testbed discriminates poorly: every arm ends between 47.7 and 56.9% macro-F1 at every level (majority-class predictor 37.5%, a balanced random predictor about 50%), all arms fit the training condition

Table D.18: Paderborn, real-damage bearings, 1500 rpm → 900 rpm. *Disjoint*: training on KA04, KA16, KI04, KI14 and evaluation on KA15, KA22, KI17, KI18, KI21. *Shared*: training and evaluation on KA04, KA16, KI04, KI14 (different recordings). Upper block: macro-F1 (%) at the final epoch, mean over 5 seeds (majority-class macro-F1: 37.5% disjoint, 33.3% shared). Lower block: paired differences under final-epoch reporting, 95% t-interval and seeds (of 5) with a positive difference. "–": arm not run in that setting. Earlier noise protocol (training and evaluation noise shared by index); see Section 4.1 for the independent-noise re-run.

	Disjoint bearings			Shared bearings		
Arm / contrast	0 dB	−2 dB	−6 dB	0 dB	−2 dB	−6 dB
BM3-frozen	51.9	50.3	47.7	84.5	85.8	86.6
frozen−all	54.2	51.3	47.8	–	–	–
S4D	55.6	53.9	51.1	76.1	77.3	77.9
frozen−gate	56.9	54.9	51.1	75.5	78.1	81.3
frozen+add	56.9	54.8	51.0	–	–	–
BM3-frozen − S4D	−3.8 [−11.9, 4.3] (1/5)	−3.6 [−12.6, 5.4] (2/5)	−3.4 [−8.9, 2.1] (1/5)	+8.4 [−1.3, 18.1] (4/5)	+8.5 [0.2, 16.8] (5/5)	+8.7 [4.3, 13.0] (5/5)
BM3-frozen − frozen−gate	−5.0 [−8.8, −1.3] (0/5)	−4.6 [−9.2, 0.1] (0/5)	−3.3 [−7.1, 0.4] (0/5)	+9.0 [5.2, 12.8] (5/5)	+7.7 [5.0, 10.4] (5/5)	+5.3 [2.4, 8.2] (5/5)
frozen−all − S4D	−1.4 [−4.0, 1.2] (1/5)	−2.5 [−5.3, 0.2] (1/5)	−3.3 [−5.8, −0.8] (0/5)	–	–	–
frozen+add − frozen−gate	0.0 [−0.9, 0.8] (2/5)	0.0 [−0.4, 0.3] (3/5)	−0.1 [−0.4, 0.2] (1/5)	–	–	–

(the gated arms reach the lowest training loss), and none transfers far above the level reached after the first epoch. Second, where differences are resolved they run against the Mamba-3 arms, which reach lower training loss and transfer slightly worse. In the disjoint setting, therefore, the block's advantage under matched noise and drift is not observed; we do not know whether this reflects a property of the PU transition (speed reduction to 900 rpm with disjoint bearings), the protocol transferred unchanged from XJTU-SY, or a limit of the XJTU-SY result itself, which the shared-bearing setting below separates.

Shared bearings. With bearing identity shared between training and evaluation, the same condition pair becomes transferable: every arm ends between 75.5 and 86.6% at every level, and the pre-specified transferability gate (5-seed mean $\geq 65\%$ with one-sided 95% lower bound $> 60\%$ at two of three levels for at least one arm) is passed by all three arms at all three levels (smallest lower bound 71.3%). Within this setting both effects replicate under the pre-specified rule (Table D.18, right). BM3-frozen exceeds S4D by $+8.4$, $+8.5$, $+8.7$ pp at 0, -2, -6 dB, with intervals excluding zero at -2 and -6 dB; removing the gate costs $+9.0$, $+7.7$, $+5.3$ pp, with intervals excluding zero at all three levels and in every seed; no level runs in the opposite direction. The gateless arm returns to the level of S4D (frozen$-$gate $-$ S4D: -0.6, $+0.8$, $+3.4$ pp, all intervals including zero), as on XJTU-SY. Under best-epoch reporting the freeze contrast is resolved at all three levels ($+6.9$, $+10.8$, $+13.7$ pp) and gate removal only at -6 dB ($+1.5$, $+3.7$, $+6.4$ pp). The cost of gate removal reaches the 8 pp threshold used on XJTU-SY at one level only; we compare directions, not magnitudes, across datasets.

Read together, the two protocols give different results: with shared bearings every arm, S4D included, ends 18.6–38.8 pp higher than with disjoint bearings (independent seeds; per-arm Welch intervals all exclude zero), and the block's advantage and the cost of gate removal reappear. Because the evaluation bearings, class balance and sample size change together, this does not isolate bearing overlap as the cause. With two bearings per class in the shared setting, a bearing's identity also determines its label, so a model can succeed partly by recognising bearing-specific signatures. XJTU-SY differs from both Paderborn settings: its training and evaluation bearings are disjoint in each direction, and the block transfers there. The shared-setting contrasts were re-run under independent noise (Section 4.1); *frozen$-$all* was not run on Paderborn.

Appendix D.9. Sensitivity to the reporting rule (earlier protocol)

The main tables use the final epoch of the 50-epoch budget (Section 3.3). The original grids recorded instead the best epoch on the evaluation condition, and their pre-specified thresholds were written for that rule. To measure how much the rule shapes the conclusions, every arm of the chain was re-run over the adjudication band with per-epoch evaluation, using the original training code and the same evaluation fingerprint: the four arms BM3-frozen, S4D, frozen−gate and S4D+gate (60 cells), the BM3 arm alongside the layered freezes (Appendix D.2), and the capacity control and the two other single-component removals, S4D-wide, frozen−conv and frozen−A (45 cells, pre-specified with a rule for changes of sign). The re-runs used a different PyTorch/Triton build from the original grids; their best-epoch values reproduce the original arm-by-condition means to within −0.6 to +1.4 pp across the eight re-run arms (the largest deviation is in the high-variance frozen−gate arm at −2 dB).

Table D.19: Macro-F1 (%), mean over 5 seeds, under best-epoch and final-epoch reporting in the re-runs, for every arm of the chain on the adjudication band. Earlier noise protocol (training and evaluation noise shared by index); see Section 4.1 for the independent-noise re-run.

	0 dB		−2 dB		−6 dB	
Arm	best	final	best	final	best	final
BM3	95.6	86.9	95.8	88.3	88.6	83.3
BM3-frozen	90.6	83.1	92.3	87.3	85.7	83.1
S4D	75.9	67.6	75.1	67.5	68.6	66.7
S4D-wide	77.8	67.3	73.3	66.5	69.0	67.1
frozen−gate	77.7	74.0	73.1	71.7	61.9	60.5
frozen−conv	97.1	96.3	95.4	94.8	77.9	77.2
frozen−A	92.5	85.1	95.6	89.9	90.4	87.9
S4D+gate	80.9	71.5	79.9	74.3	69.2	66.6

Best-epoch values exceed final-epoch values by 0.6–10.6 pp across the 24 arm-by-level means, so best-epoch absolute levels are optimistic. The reduction is of similar size for the arms being contrasted (e.g., BM3-frozen −7.4, −5.0, −2.6 pp and S4D −8.3, −7.6, −1.9 pp at 0, −2, −6 dB), and the contrasts that carry the argument are correspondingly stable (Table D.20).

Table D.20: Paired differences under final-epoch reporting, in percentage points: mean, 95% t-interval over seeds, and number of seeds (of 5) with a positive difference. Earlier noise protocol (training and evaluation noise shared by index); see Section 4.1 for the independent-noise re-run.

Contrast	0 dB	−2 dB	−6 dB
BM3-frozen − S4D	+15.5 [4.3, 26.8] (5/5)	+19.8 [12.0, 27.7] (5/5)	+16.4 [4.1, 28.7] (4/5)
BM3-frozen − frozen−gate	+9.1 [2.9, 15.3] (5/5)	+15.6 [−0.8, 32.0] (4/5)	+22.6 [5.8, 39.4] (5/5)
frozen−gate − S4D	+6.4 [−2.9, 15.8] (4/5)	+4.2 [−14.5, 22.9] (3/5)	−6.2 [−23.5, 11.2] (1/5)
S4D+gate − S4D	+3.9 [−0.8, 8.6] (4/5)	+6.8 [2.4, 11.1] (5/5)	−0.1 [−4.0, 3.9] (2/5)

Selectivity. The frozen block's advantage over S4D is +15.5, +19.8 and +16.4 pp (and +21.9, +24.0, +21.2 pp for the fully frozen scan, Appendix D.2), versus +14.6, +17.2 and +17.1 pp under best-epoch reporting in the re-run. It remains positive in 5/5, 5/5 and 4/5 seeds, with intervals excluding zero at every level. *Gate.* Removing the gate costs 9.1, 15.6 and 22.6 pp (12.8 to 23.8 pp under best-epoch reporting in the re-run), above the 8 pp pre-specified threshold at every level; the interval at −2 dB includes zero ($[-0.8, 32.0]$ pp, 4/5 seeds) because of the large seed variance of the gateless arm (standard deviation 14.6 pp at −2 dB). *Gateless versus S4D.* Under final-epoch reporting the gateless arm is not below S4D at 0 and −2 dB (point estimates +6.4 and +4.2 pp) and is below it at −6 dB (−6.2 pp); all three intervals include zero. The statement that gate removal "falls below S4D" is therefore supported only as a point-estimate observation at the deepest level, and we phrase it as "to or below" elsewhere. *Graft.* The recovery ratio is 0.249, 0.341 and −0.003 per level and $R = 0.196$ on average (best-epoch re-run: 0.220), again in the pre-specified "insufficient" bin on the point estimate. The seed-resampled 95% interval is $[0.09, 0.31]$ and so reaches the 0.3 boundary: with five seeds the data support "far below confirmation (0.7) and probably below 0.3", not a precise value of R. The graft gain over S4D is resolved at −2 dB only (+6.8 pp, $[2.4, 11.1]$) and is indistinguishable from zero at −6 dB.

Remaining arms. By the rule pre-specified for the 45-cell grid, no arm changes sign between the two rules at any level where both values exceed 1 pp, so all main-table entries are reported under the final-epoch rule with unchanged wording. The capacity control holds: BM3-frozen exceeds S4D-

wide by +15.9, +20.8, +16.0 pp (intervals excluding zero at all three levels), and S4D-wide − S4D is −0.4, −1.0, +0.4 pp (intervals including zero). For the stem the final-epoch rule sharpens the regime trade-off (intervals at 0 and −2 dB now exclude zero) without reversing it.

Because the final epoch of the same evaluation condition has been inspected repeatedly during this study, last-epoch values are also not a fresh confirmatory test. The reporting rule therefore changes absolute levels and the strength of the gateless-versus-S4D statement, but not the ordering BM3-frozen > {S4D+gate, frozen−gate, S4D}, the size of the frozen block's advantage over S4D, or the verdicts of the pre-specified gate and graft rules.

Appendix D.10. Per-class recall (earlier protocol)

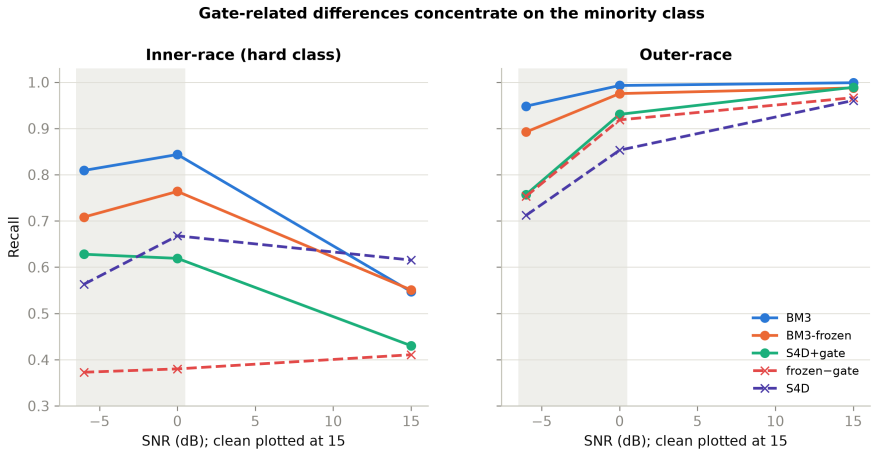


Figure D.5: Per-class recall (inner-race, IR, and outer-race, OR) as a function of noise depth, split by gate presence. Gated arms lift IR recall under matched noise; gateless arms let it deteriorate. Earlier noise protocol (training and evaluation noise shared by index).

Per-class analysis (a separate 75-cell grid at 20 epochs with best-epoch selection on the evaluation condition: 5 arms × {clean, 0, −6 dB} × 5 seeds, per-sample predictions dumped and re-verified against reported metrics cell-by-cell): the inner-race class is hard from the outset (clean recall 0.41–0.62 vs. 0.96–1.00 for outer-race). Its trajectory under noise is governed by gate presence: in all three gated arms IR recall at −6 dB exceeds its clean value (e.g., 0.55 → 0.81 for BM3), while in both gateless arms it is below its clean value (frozen−gate: 0.41 → 0.37; S4D: 0.62 → 0.56, non-monotone, rising

to 0.67 at 0 dB) (Figure D.5). Both ablation directions hit IR hardest—gate removal costs IR 2.3–3.8× more F1 than OR at every level (F1 from predictions pooled over seeds), and the graft's noise-time gains also concentrate on IR (−6 dB: +6.8 pp IR vs. +4.2 pp OR). Because macro-F1 averages the two classes, the arm-level robustness gaps of Appendix D.1–Appendix D.4 are carried substantially by the hard class. The gate-related differences are therefore class-concentrated rather than uniform, and fall on the class with the lowest recall. We did not measure classification margins, so we do not describe IR as a "low-margin" class. Because this grid is separate from the main 50-epoch grid and the IR class comes from a single training bearing (Section 3.1), these results are descriptive and do not decompose the main-grid effects.

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